



National Bank of Georgia

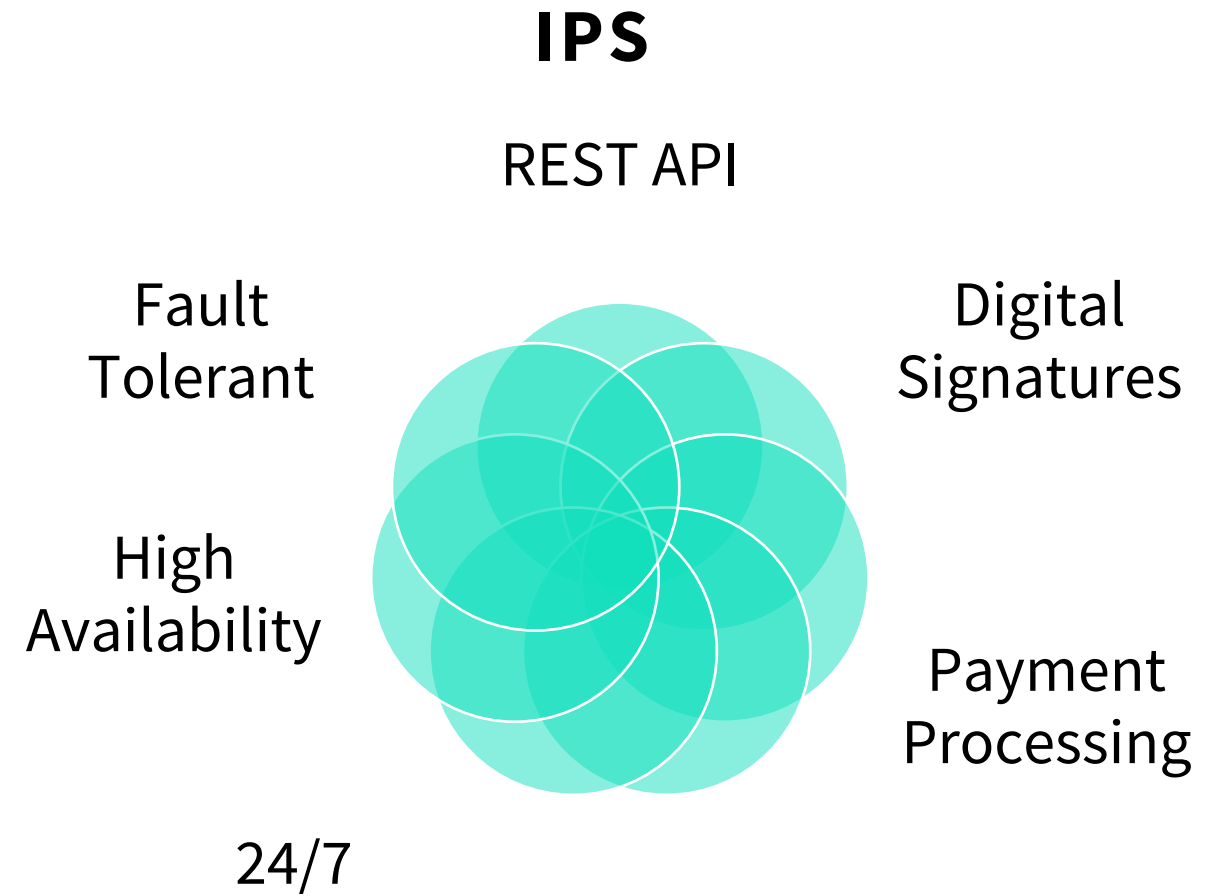
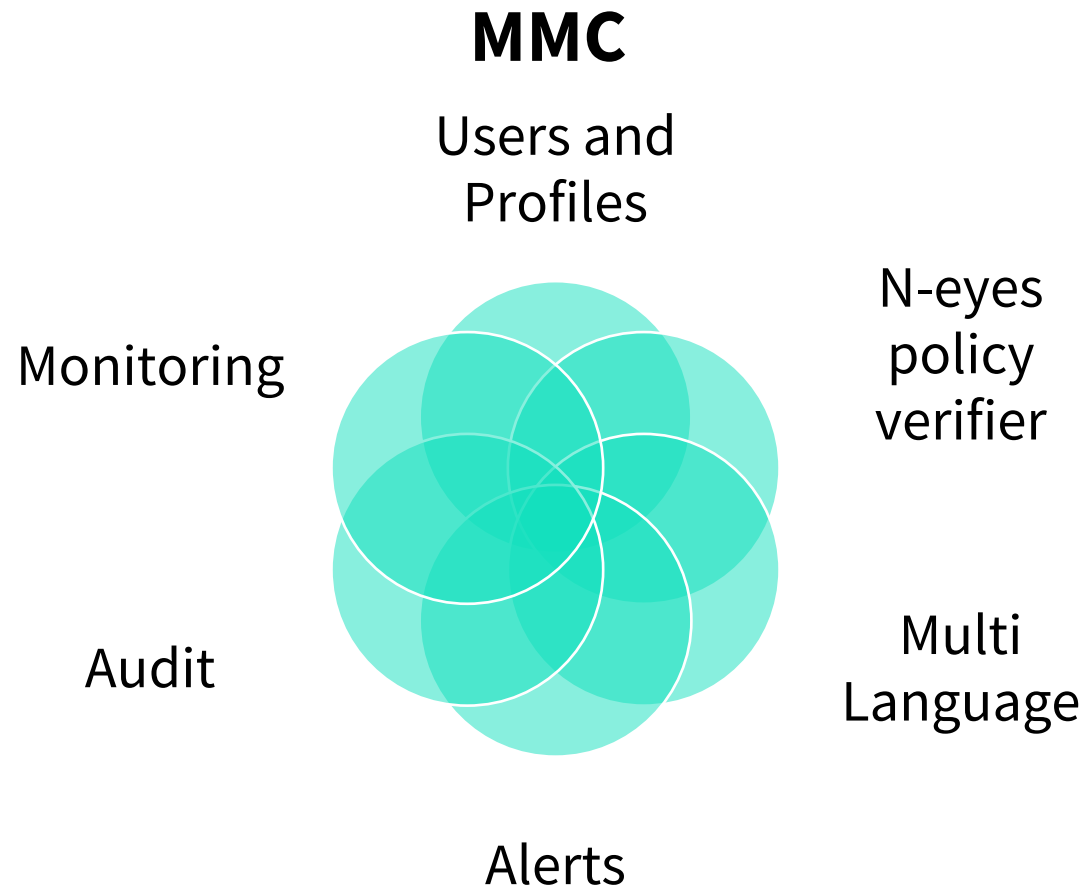
IPS/Proxy Participants Training

General concepts about IPS

Instant Payments Definition

- Solution for retail electronic payments
 - » Instant compensation for interbank payment messages
 - » Providing to the Sender Participant with the confirmation that the transferred funds have been made available to the beneficiary
 - » Guarantees the final settlement of the funds to the Receiver Participant
 - » The beneficiary must have the funds immediately available to use them.
- Real-time/ Instant Credit Transfer (ICT) payments from payer to payee in less than 7 seconds
 - » Process all types of messages
 - › ICT and additional messages (R-messages), Request to Pay, Payment Initiation
 - » Low processing latencies
- Operates 24/7/365 with 99.99% availability

IPS MMC and IPS Basic Feature



Montran Central Instant Payments System

High Availability and Resilience

Continuous 24 x 7 processing

Maintenance or upgrade with no downtime

- Individual IPS Processor nodes can be stopped and restarted without service loss

Multiple active nodes on production site

Dynamic cluster automatic re-configuration

Disaster recovery site for contingency

- Real-time replication between sites

Montran Central Instant Payments System

Additional Overlay Services

Dispute Management System

Overlay Services – other payment types, Request to Pay

Proxy Database System

Montran Proxy Solution

Overview

Use Case

- Payment initiation using secondary identifiers (phone number, email address) instead of the complete beneficiary information (full name, account number, servicing bank)

Scope

- Maintain a centralized database that matches secondary account identifiers to corresponding IBAN account numbers
- Respond to incoming lookup inquiries
- Use a standard messaging scheme based on ISO20022

Access and Integration

- Write access to proxy information is controlled and restricted to participants only
- Integration with (instant) payments systems
 - » Out-of-flow - secondary identifiers are resolved before the payment is submitted; payment messages contain IBAN account numbers

Montran Proxy Solution

Key Features

Individually running service

Available 24 x 7 x 365

High level of resilience

Ensures security, authenticity and confidentiality of data

Supports out-of-flow operation, via API provided to Participants

Participant Management – User and Profile Management

- Each IPS User has to be registered by Operator.
- The User can be added to IPS by a Participant Security Administrator using the appropriate IPS menu functions.
- Each User has a User Profile that controls the menu it can see.

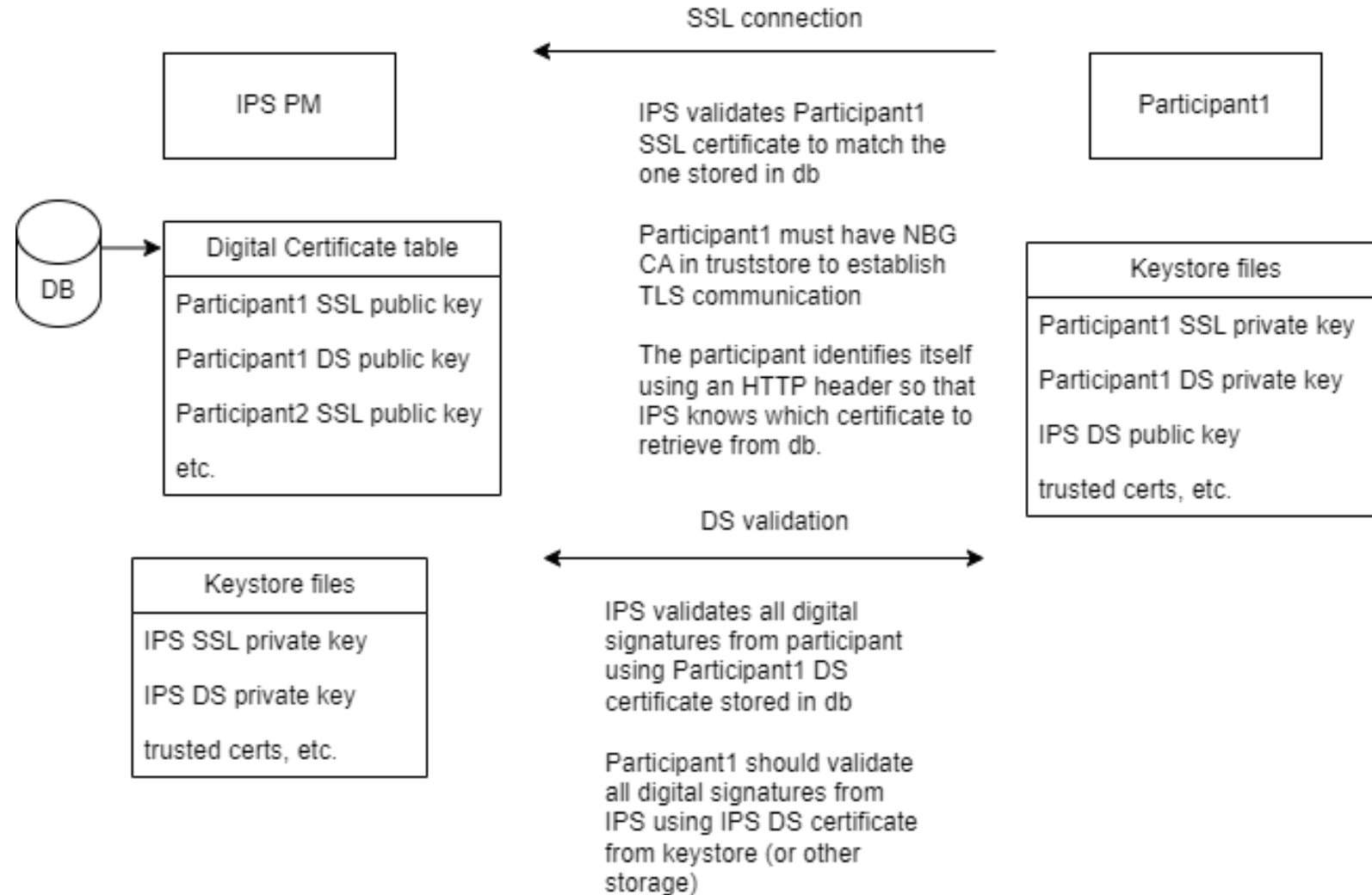
Security Schema – Access to MMC

- Access Control Prerequisites:
 - » Username
 - » Password
 - » User Profile
 - » One-time-password (OTP)
- **Hands-on:** login using your initial user and password

Security Schema – Access to IPS REST API

- Participants connect via VPN for submission and receipt of payment instructions using the REST API
- Access via VPN to the MMC which has a web-based interface for interactive and maintenance functions
- VPN connection is private and secure, all exchanged messages are signed
- Digital certificates management
 - » TLS certificates
 - » Digital signatures - ECDSA

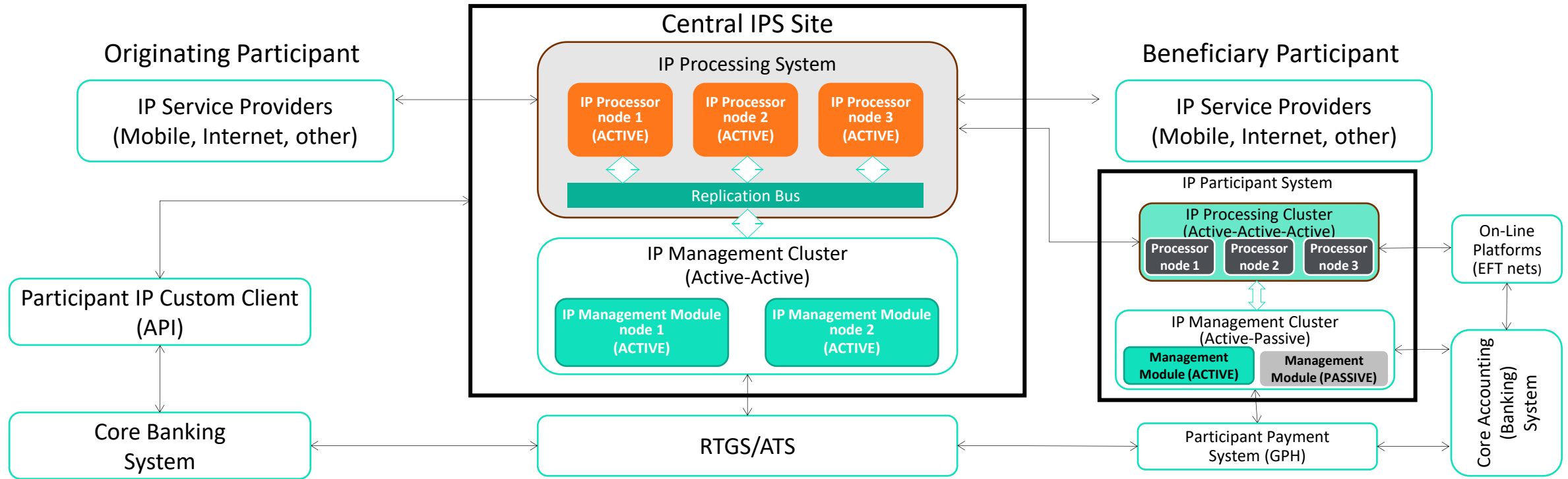
Security Schema – Access to IPS REST API



IPS Straight-Through-Processing (STP) for Financial Institutions

Montran Central Instant Payments System

Instant Payments End-to-End Solution



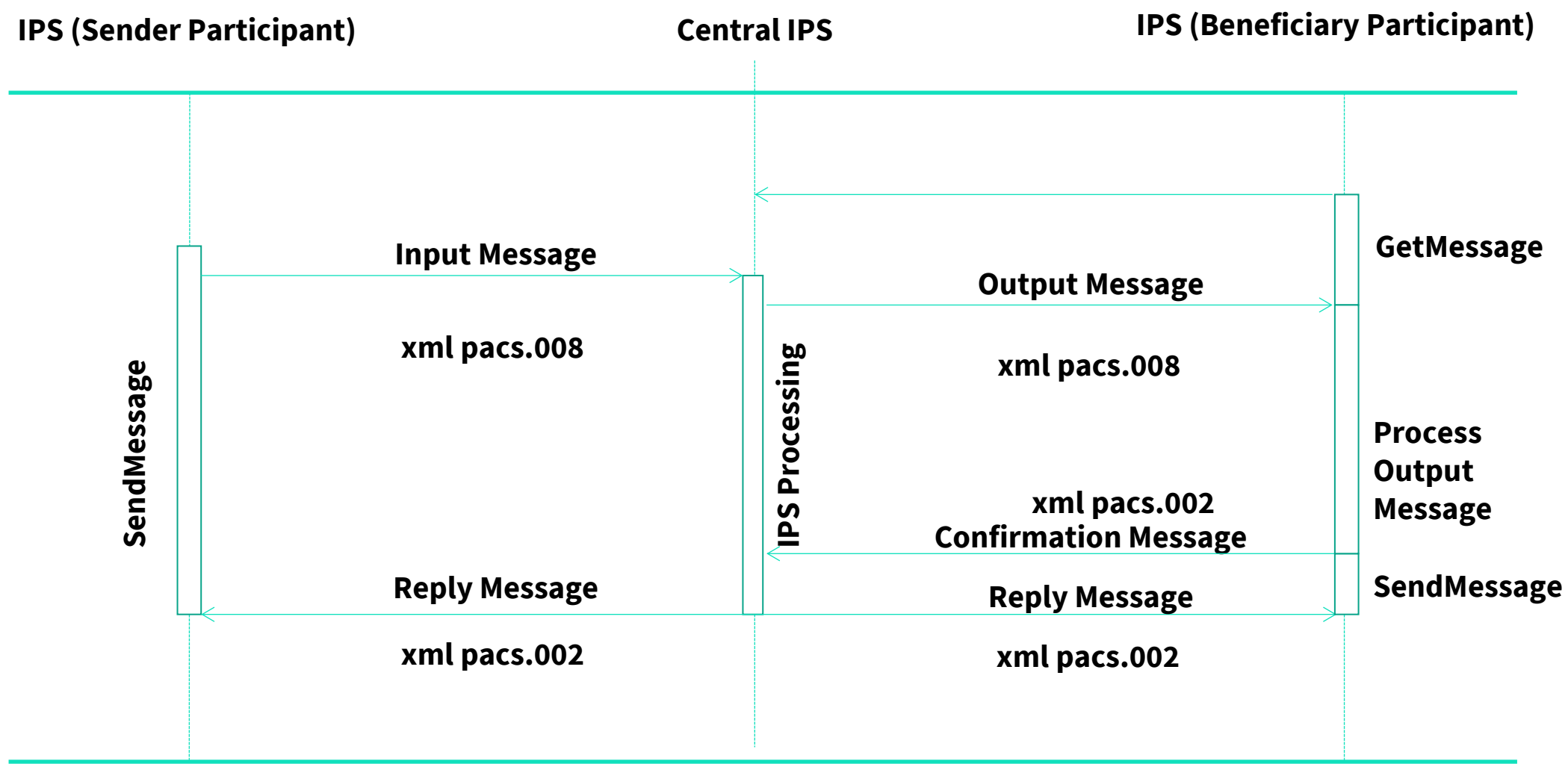
STP Interface

- The STP interface is developed using the HTTPS client-server protocol, through which the Participants' STP applications constantly request from the central IPS system to execute synchronous operations.
- The Customer application initializes a request and awaits receiving an answer from the central system (server).
- IPS implements a duplication-detection algorithm so that sending and re-sending of a message to IPS means that IPS processes the first received message.
- The STP communication protocol is stateless, i.e. each operation is completely independent from the others.
- The interface allows accessing a single function for the execution of a business operation.

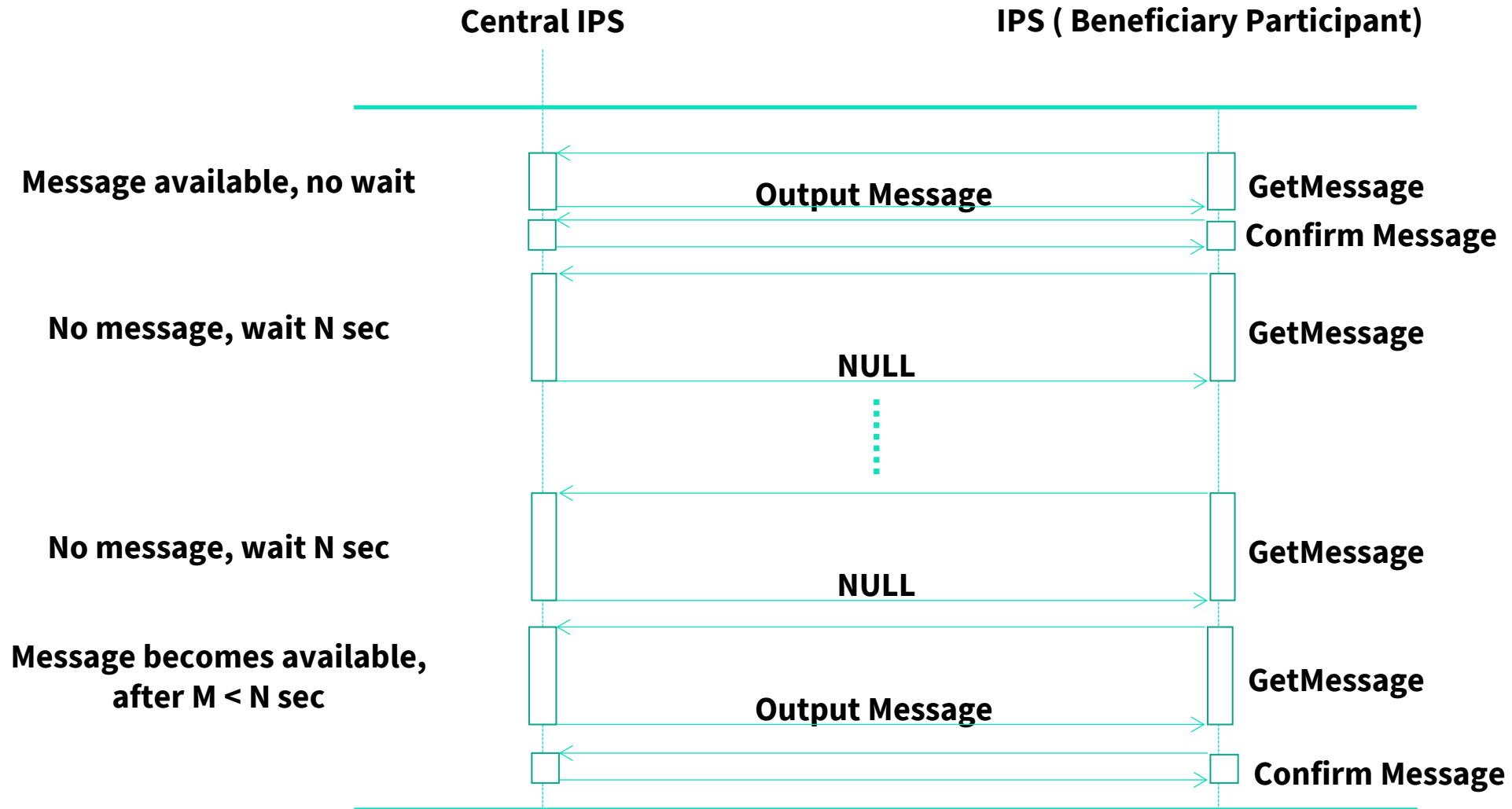
STP Interface Operations

- The STP Interface offers the following Participant operations:
 - » Send messages according to the SCT Inst schema / ISO.
 - » Receive messages (both from SCT Inst schema and additional reconciliation messages or general notifications generated by the central IPS system).
 - » Send confirmation message – pacs.002 for a received credit transfer instruction – pacs.008.
 - » Send confirmation of received message (only for messages different from pacs.008, pacs.008 confirmation is described above).
 - » Queries for: technical accounts' positions and available clearing limit for own account.

Send Messages

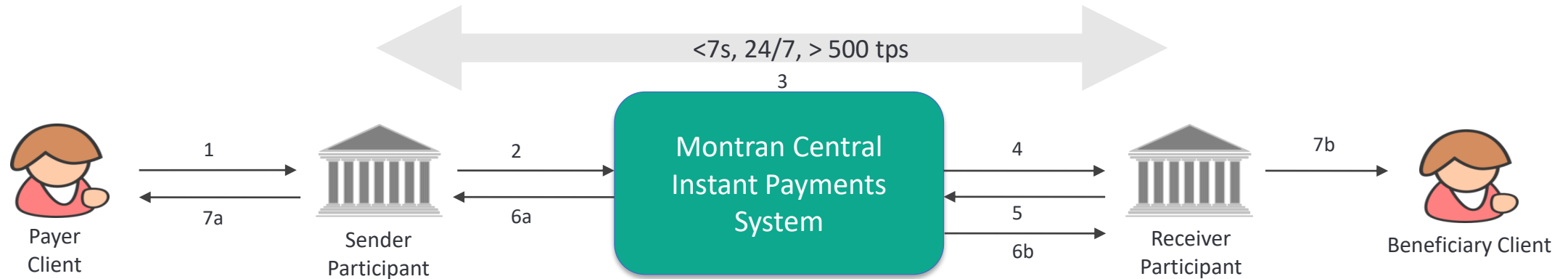


Receive Messages



Montran Central Instant Payments System

Basic Transaction Flow



1. The payer client initiates a transaction through a payment channel provided by its bank.
2. The sender participant sends a message with all payment details to the Central Instant Payments System.
3. The Central Instant Payments System validates the payment message and, if the Sender Participant has sufficient liquidity, blocks the funds to be transferred from the sender's account.

4. The payment instruction is sent to the Receiver Participant to validate the details of the beneficiary client.
5. When the Receiver Participant responds positively (the account can be credited), the Central Instant Payment System changes the status of the transaction to Final and Complete, and transfers the funds to the Receiver Participant's account, increasing its available balance.
- 6a and 6b. The sender and receiver participants are notified of the outcome of the transaction.

- 7a. The sender participant informs the payer client of the outcome of the transaction.
- 7b. The receiver participant sends a notification to the beneficiary customer of the funds credited and their availability for immediate use.

IPS Participants Implications and Solution

Instant Payments

Participant Implications

- Interfaces with the IPS Central via a REST API
 - » HTTPS over VPN
 - » Basic Java Library offered to Participants for integration with Bank solutions
- Custom Development Around Provided IPS Java Library
 - » Solutions can be time consuming
 - » Extensive Testing Requirements
 - » Must Remain up-to-date with Central Solution future enhancements - Future Proof

Instant Payments

Participant Implications

- Solutions must meet High Availability and SLA Requirements
 - » Continuous 24 x 7 x 365 processing
 - » Maximum 7 seconds end to end, from the moment the payment messages is generated up to when funds are applied and made available to the beneficiary
 - » Maintenance or upgrade with no downtime
 - » Architecture with multiple active nodes and disaster recovery
- Back Office Integration
 - » Core Banking, 24 x 7 Accounting
- Support Requirements
 - » Bank customers will immediately notice service disruption
 - » 24 x 7 x 365 helpdesk
 - » Provisions for on call staff for various levels of severity

Instant Payments

Participant Implications

- Core Banking / Cash Management System Updates
 - » Approach Vendor
 - » Relay Technical Specifications
 - » Commercial agreement
 - » Deploy changes in the CBS
 - » Wait for availability of Central Test Environment
 - » Start testing
 - » Readiness Certification

Instant Payments

Participant Implications – Performance recommendations

- Connection pooling
 - » Frequent HTTPS connections can introduce overheads
 - » Use a pooling mechanism which manages and reuses connections
- Connection parameters
 - » Socket timeout – less than the payment schema deadlines
 - » Keep-alive time – add 'Connection: keep-alive' header when using connection pooling so that the server doesn't close the connections
 - » Compression – add 'Accept-Encoding: gzip' header to enable compression

Instant Payments

Participant Implications – Performance recommendations

- Efficient processing model

- » Prioritization – payment messages (pacs.008) should be handled with more priority than other messages (reconciliations etc.)
- » Blocking/Non-blocking IO – asynchronous processing can use CPU more efficiently
- » XML parsing – recommended to use a library which doesn't load the entire model in memory but goes through it sequentially

Liquidity Management

Participants Management – Participants and Accounts

- **Participant** – payment institution that has access to the IPS
- **Technical account:**
 - » Each Participant holds a technical account for the currency configured in the system.
 - » Calculates and manages the Participant's net position consistently during a liquidity adjustment.
- **Net position – technical account balance**
 - » The algebraic total amount of payments and offsetting receipts of a Participant to the IPS, calculated following the exchange of payment instructions with other IPS participants.

Participant Requirements

- Payment schema mapping creates automatically the Participant's account
 - » Maintenance → Participant → List
 - » Maintenance → Payment Schema → List
- Account menu sets limits and holiday limit increments
 - » Maintenance → Account
- Account limit menu applies one-time changes to limits prior to a liquidity adjustment event
 - » Account menu limits remain as set prior
 - » Maintenance → Account Limit

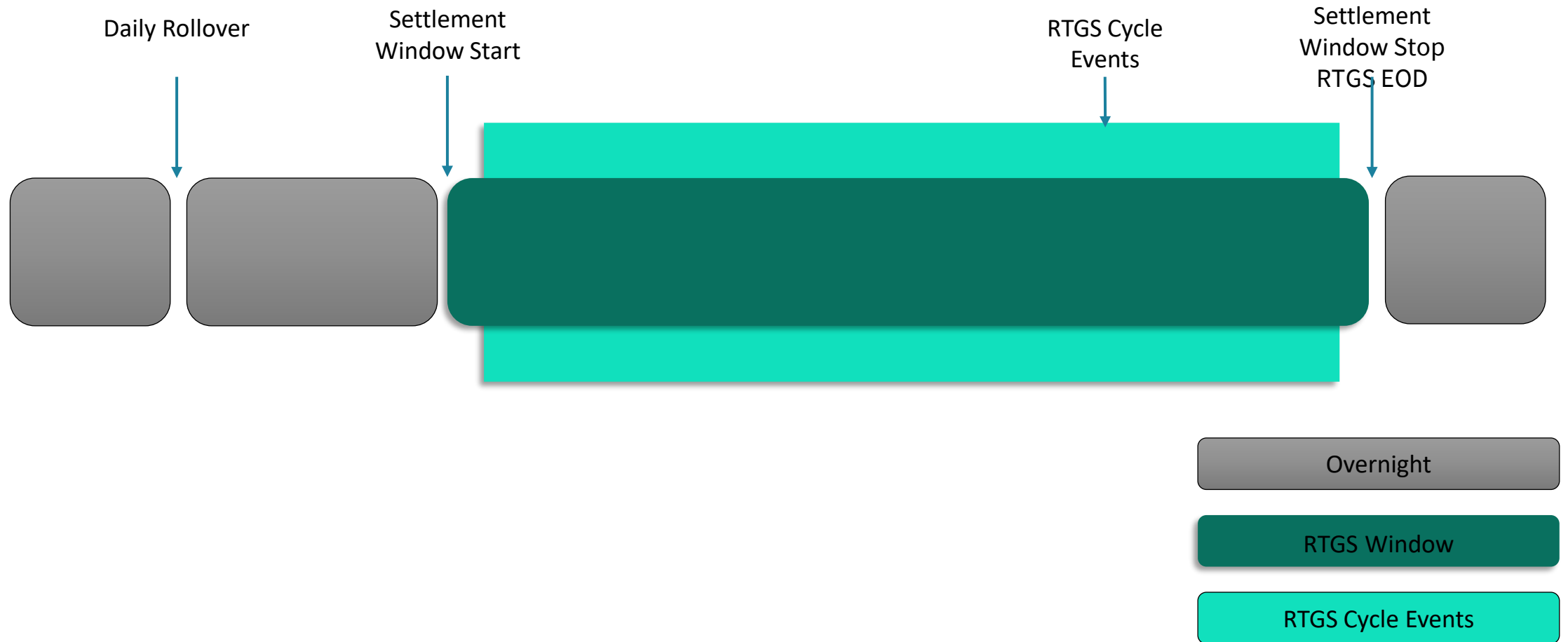
Business Calendars: IPS and RTGS

- A calendar represents one month and determines which days are business days or holidays within that month.
- Holiday days can be set anytime for future days only
- IPS handles two calendars:
 - » RTGS sync calendar - RTGS
 - » IPS internal calendar – Default

Timetables: IPS and RTGS

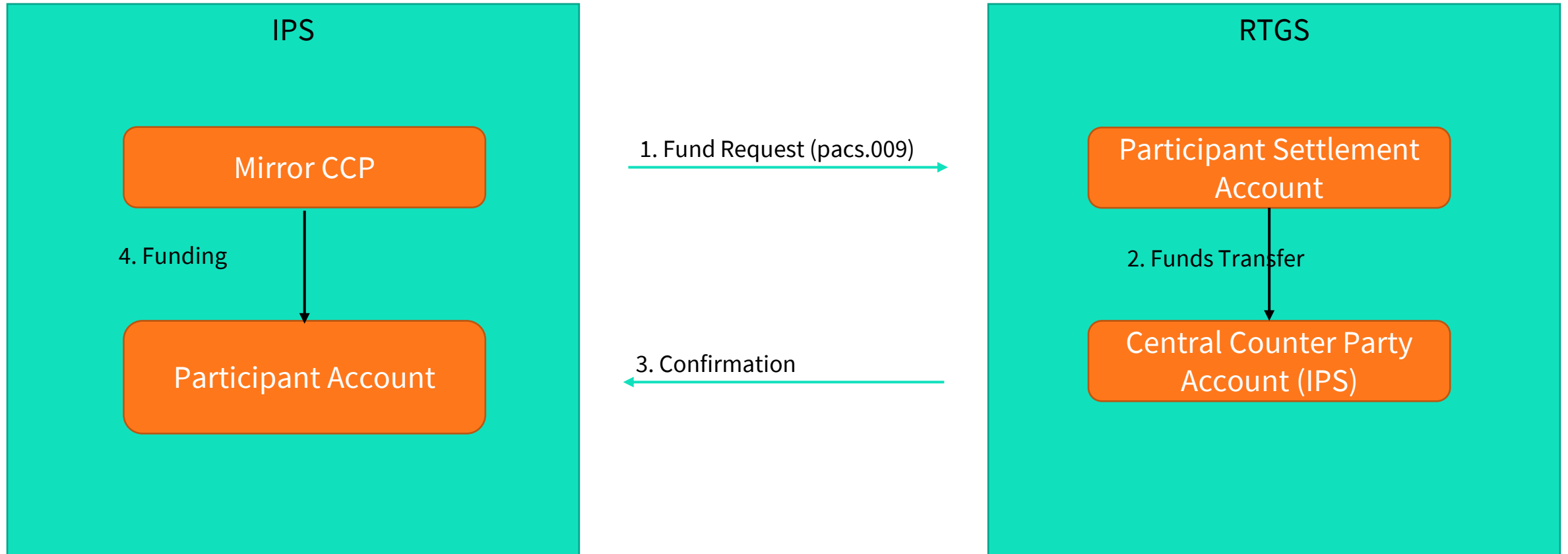
- At Daily Roll Over:
 - » Timetable advances the business day based on Calendar rules
 - » Generates reports, at set timetable event
- Settlement Window Start:
 - » Allows the execution of RTGS cycle events (settlement)
- Settlement Window Stop:
 - » Stops the interaction with RTGS
- RTGS End of Day
 - » Marks the End of Day event in the RTGS calendar

IPS Business Day



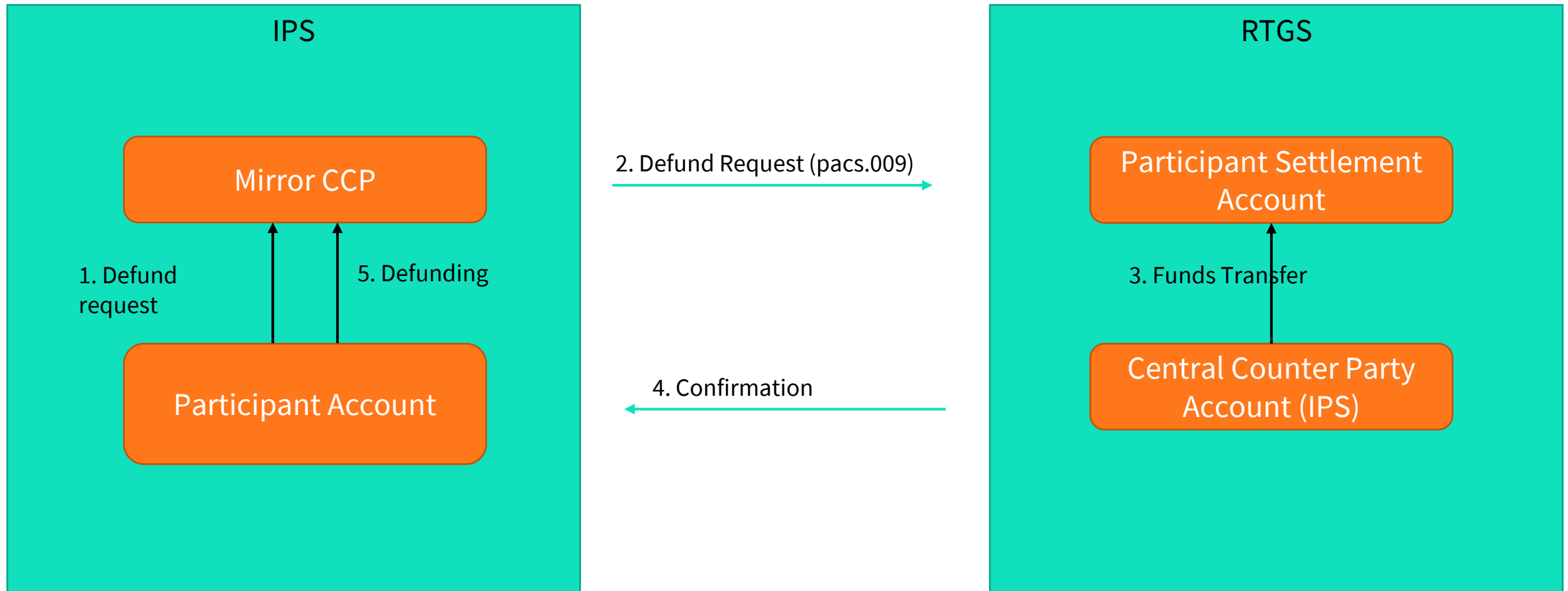
Account Management – Liquidity Management

- Fund request sent to RTGS

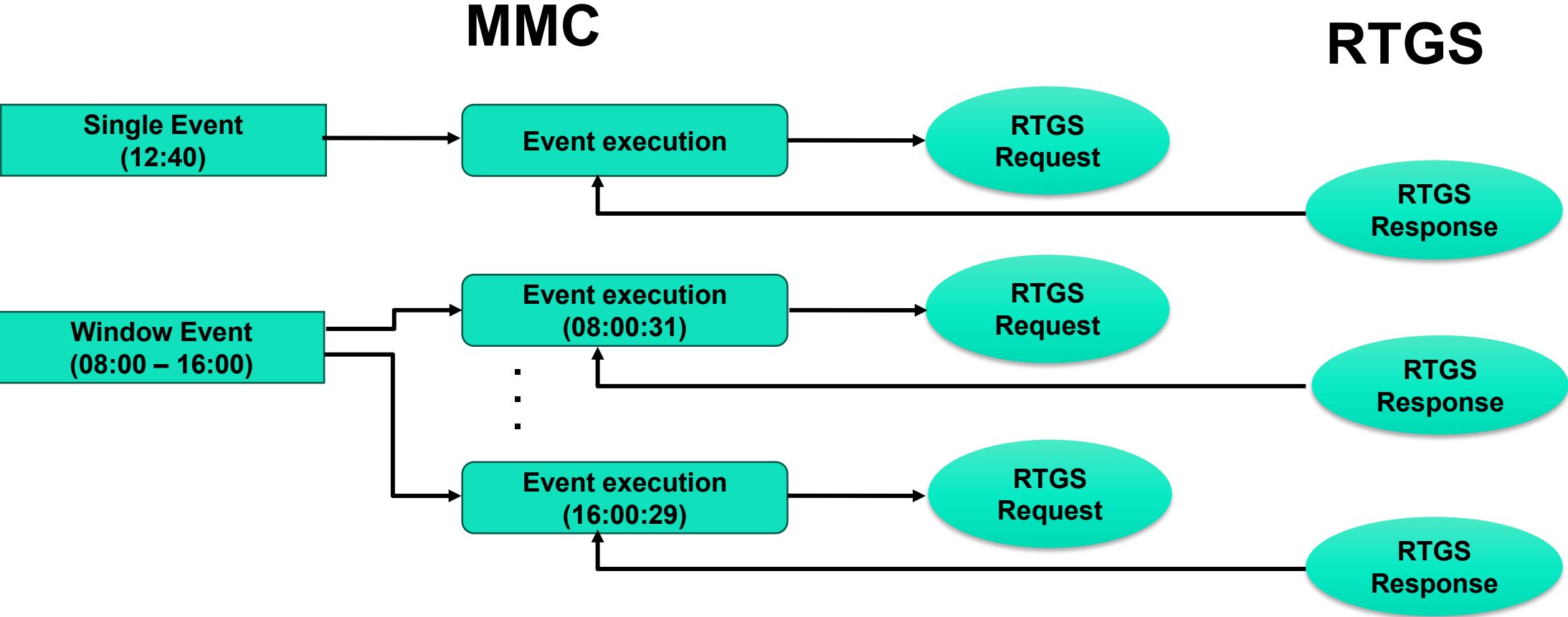


Account Management – Liquidity Management

■ Defund to RTGS



Timetable Administration Events Scheduling



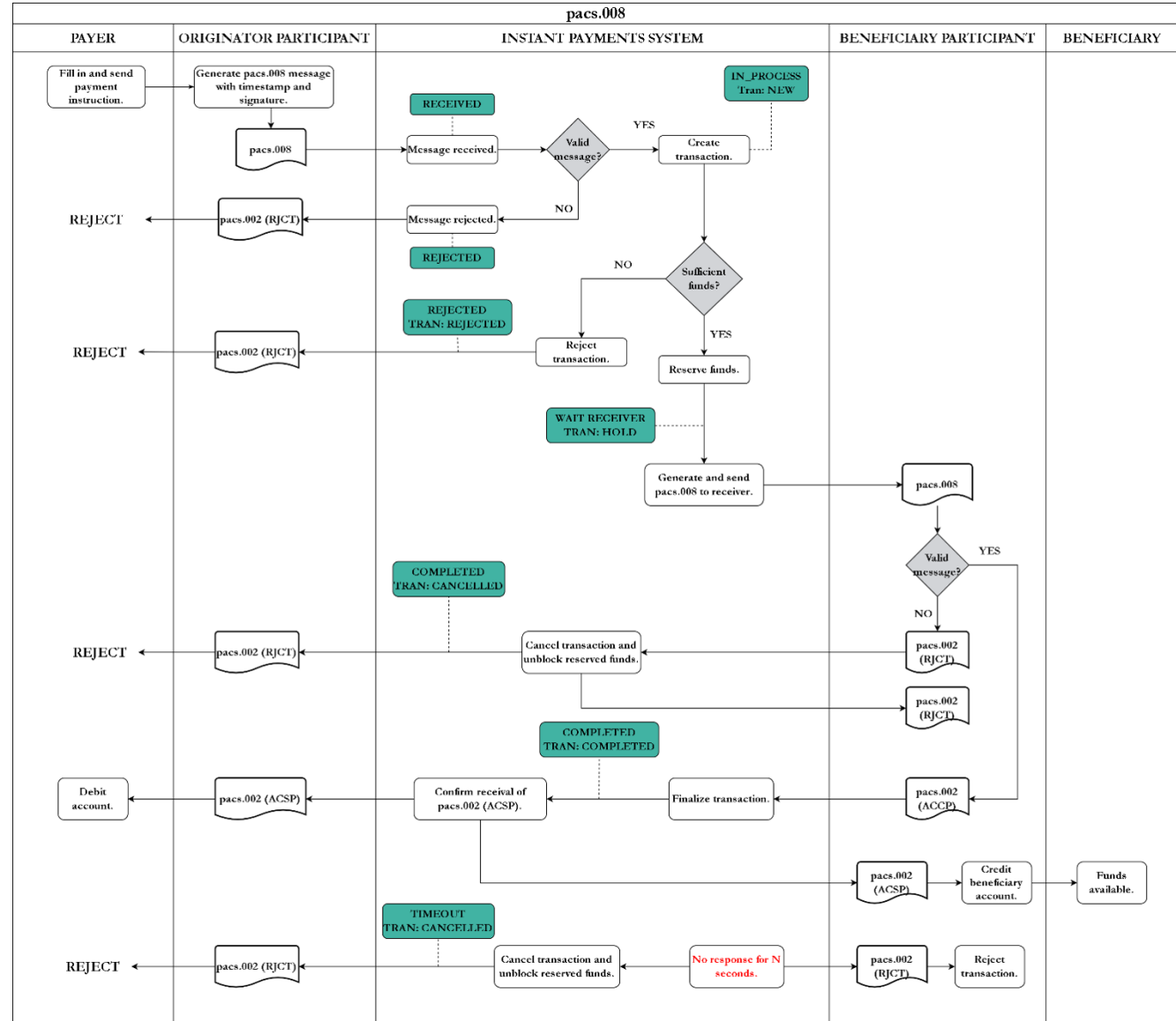
Participants – Settlement Types

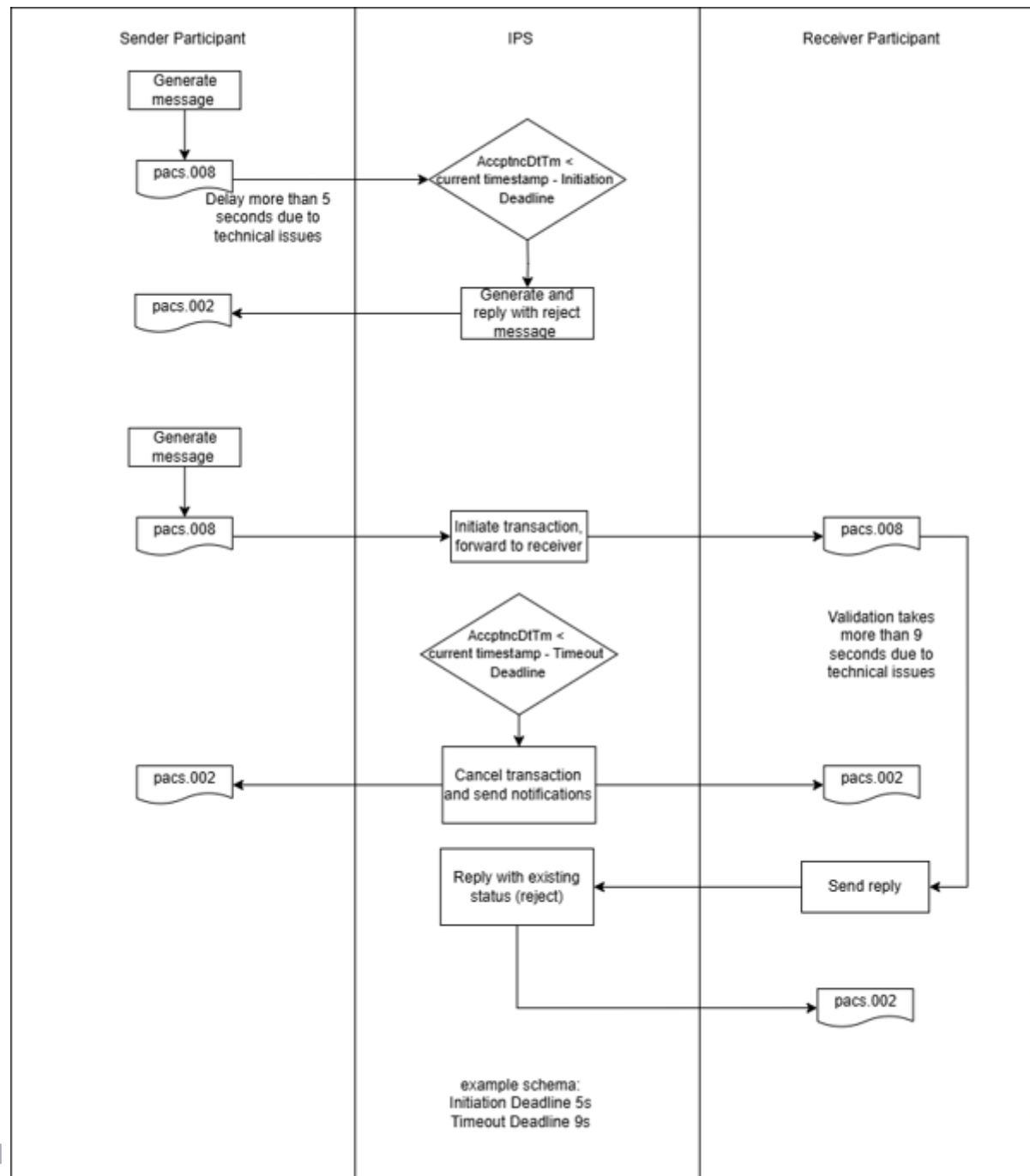
- In terms of the settlement process, IPS supports two types of settlement:
 - » **Direct Participant (DP)** – Participant having own account in the settlement system (RTGS)
 - » **Indirect Participant (IP)** – Participant for which the settlement is performed indirectly through the account of a Settlement Agent. The Settlement Agent must be a Direct Participant.

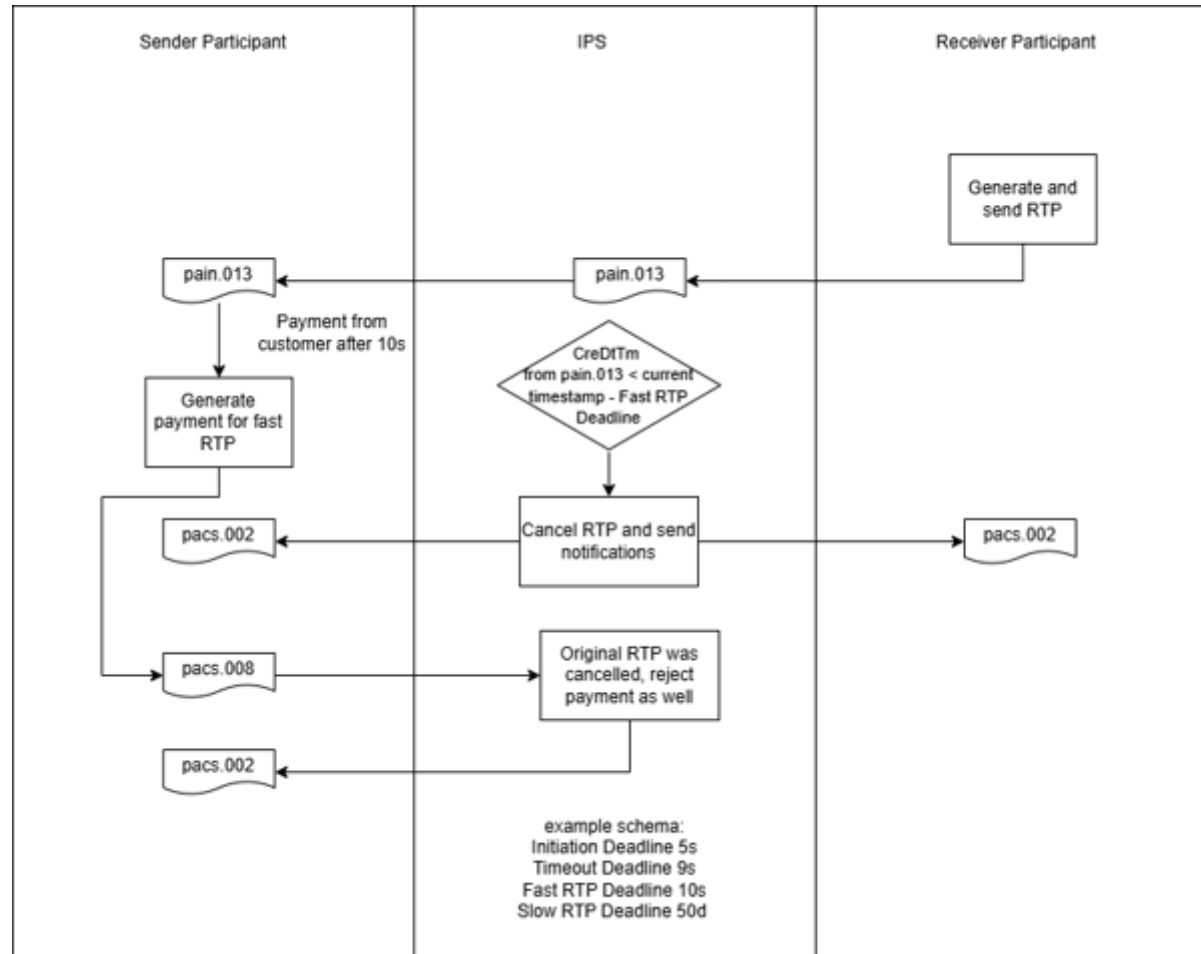
IPS Payment Flows

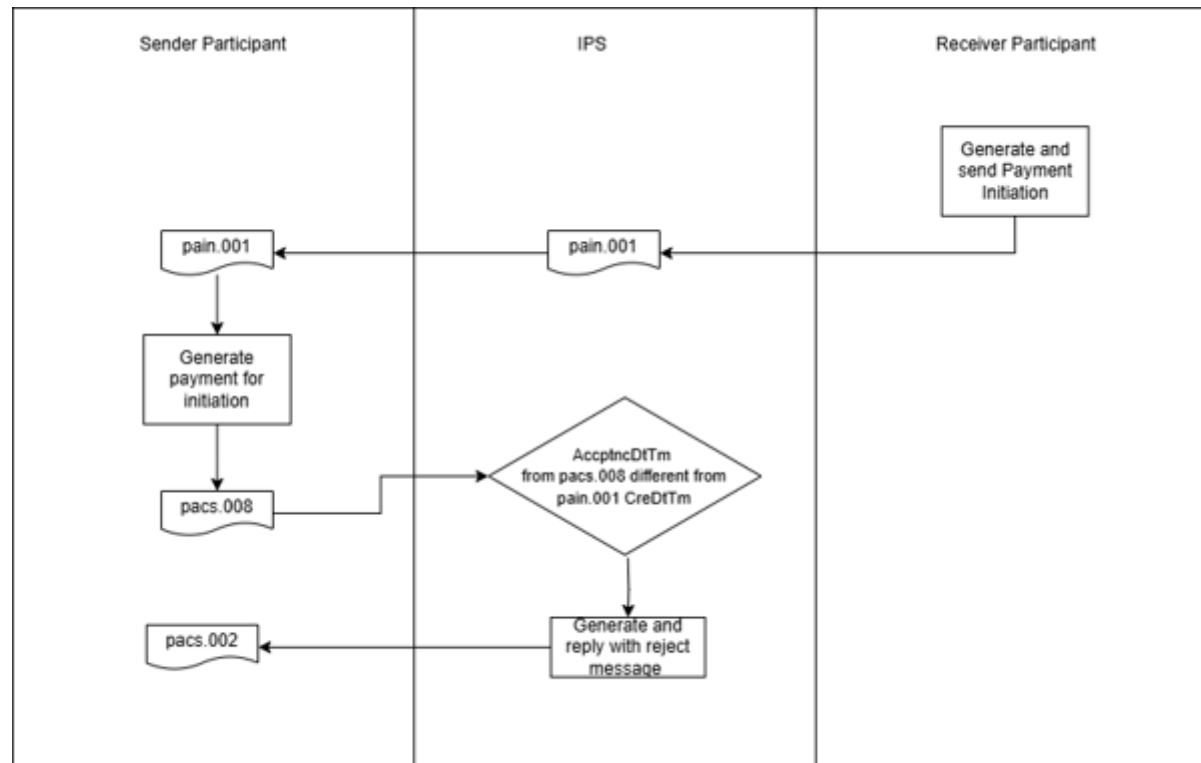
IPS Messaging – ISO 20022

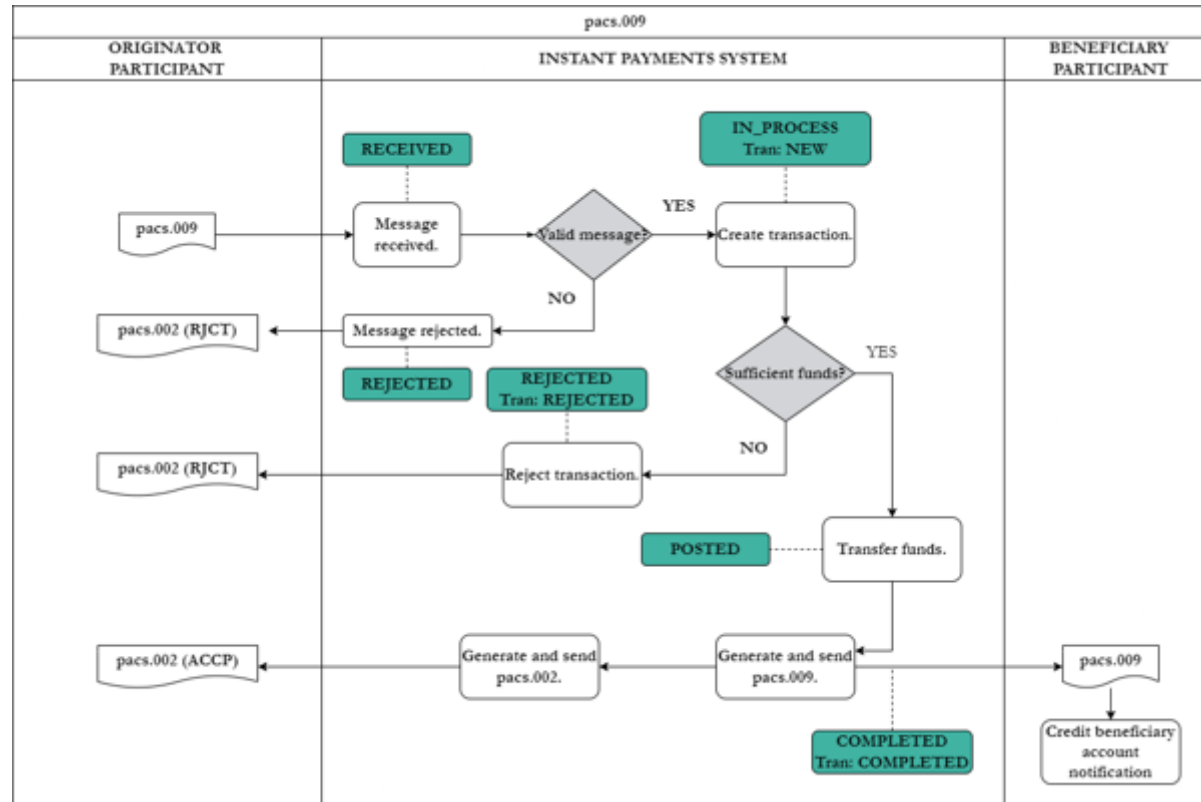
- **pacs.008** – FI to FI Customer Credit Transfer
- **pacs.009** - FI to FI Credit Transfer
- **pacs.002** – Positive and Negative Confirmation Message
- **camt.056**– Recall of an SCT Inst
- **camt.029** – Negative Response to a Recall of an ICT
- **pacs.004** – Positive Response to a Recall of an ICT
- **pacs.028** – ICT Transaction Status Investigation
- **Pain.001** – Payment initiation
- **Pain.002** – Response to payment initiation
- **pain.013** – Request to pay
- **pain.014** – negative response to request to pay
- **Pacs.055** – Request for Cancellation
- **camt.053** - Reconciliation message generated by IPS
- **positions.001** – sent by IPS to participants
- **participants.001** – sent by IPS to participants

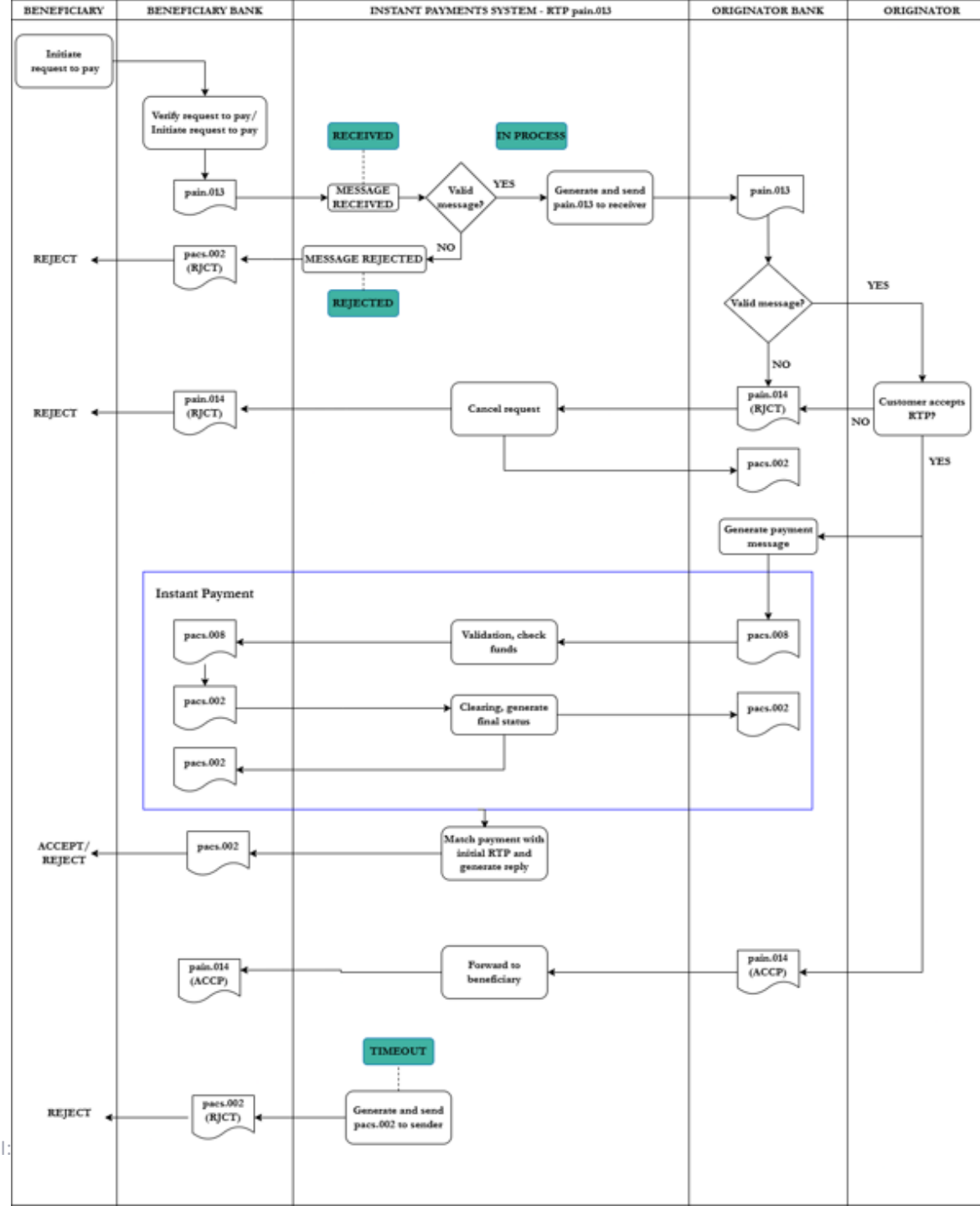


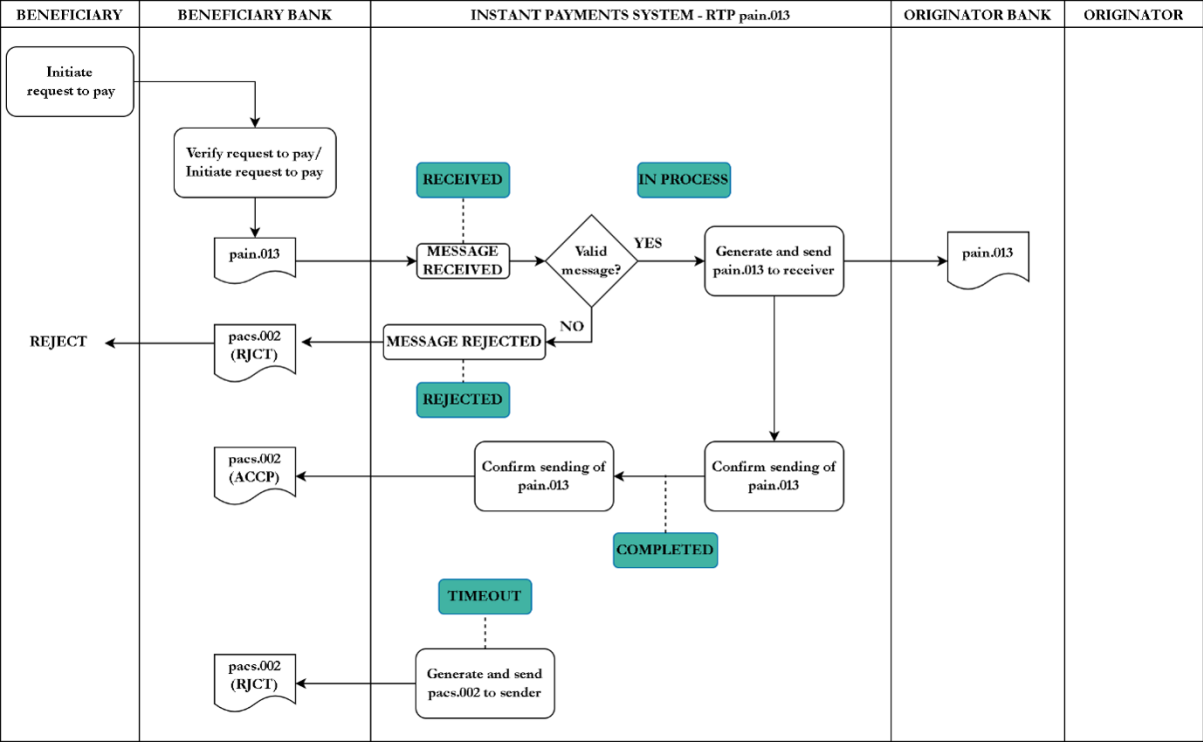


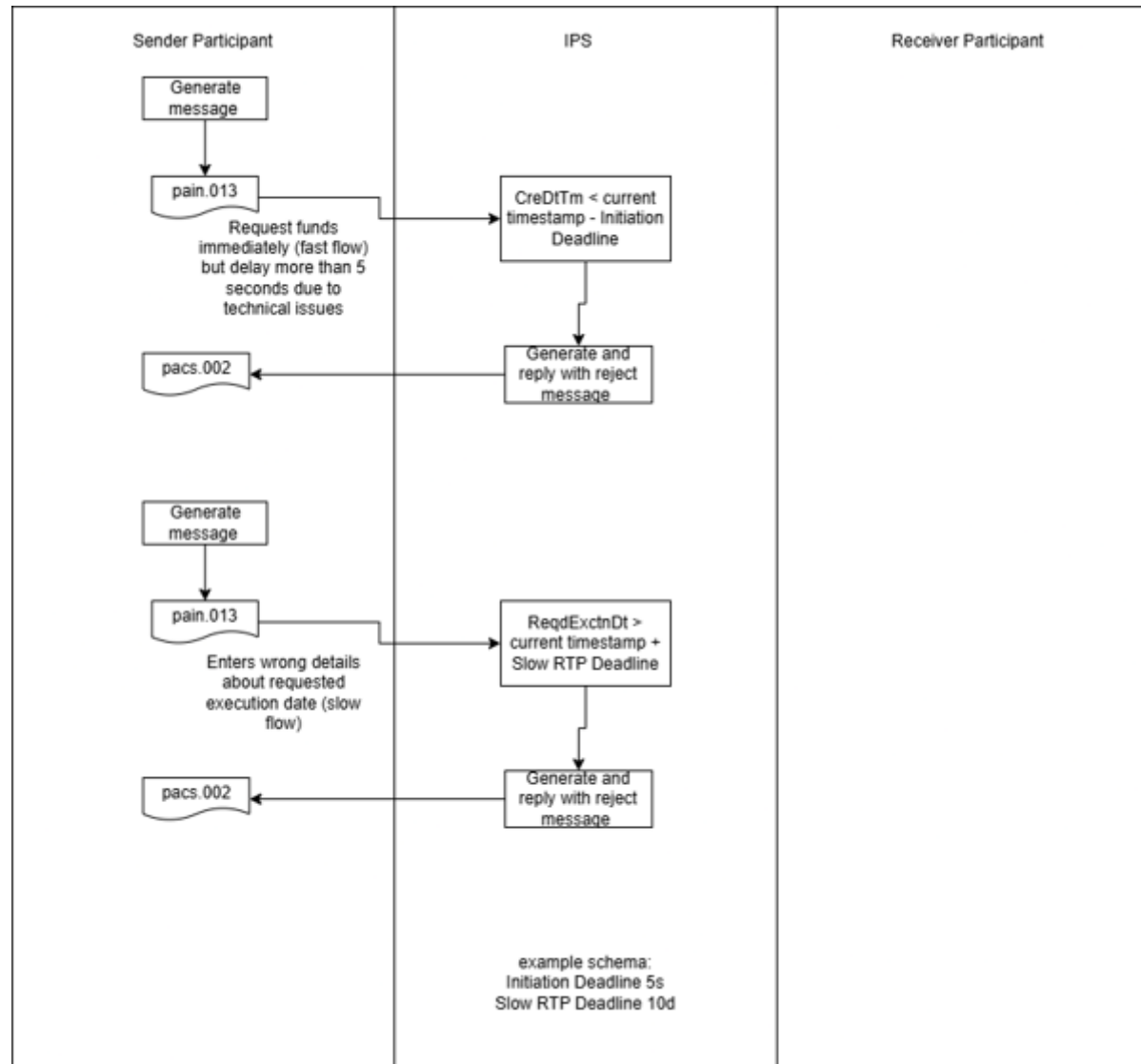


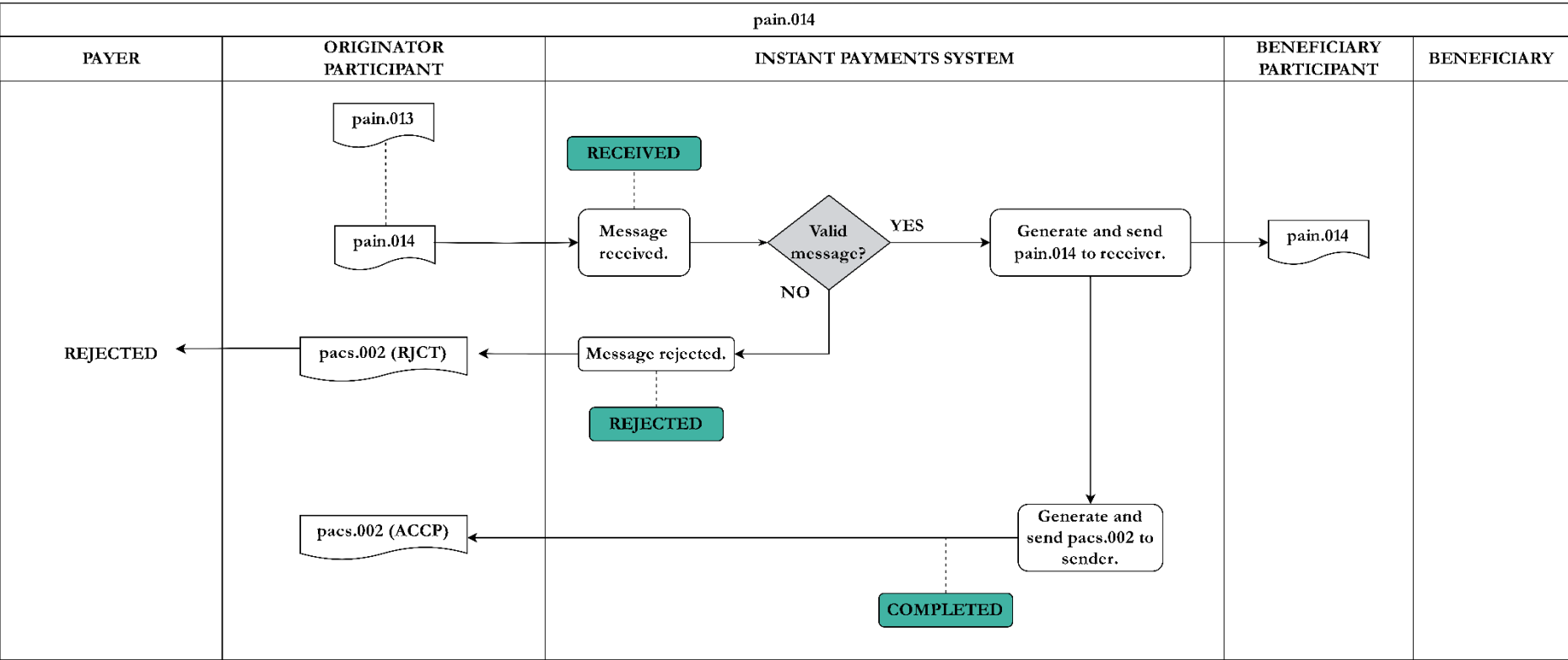


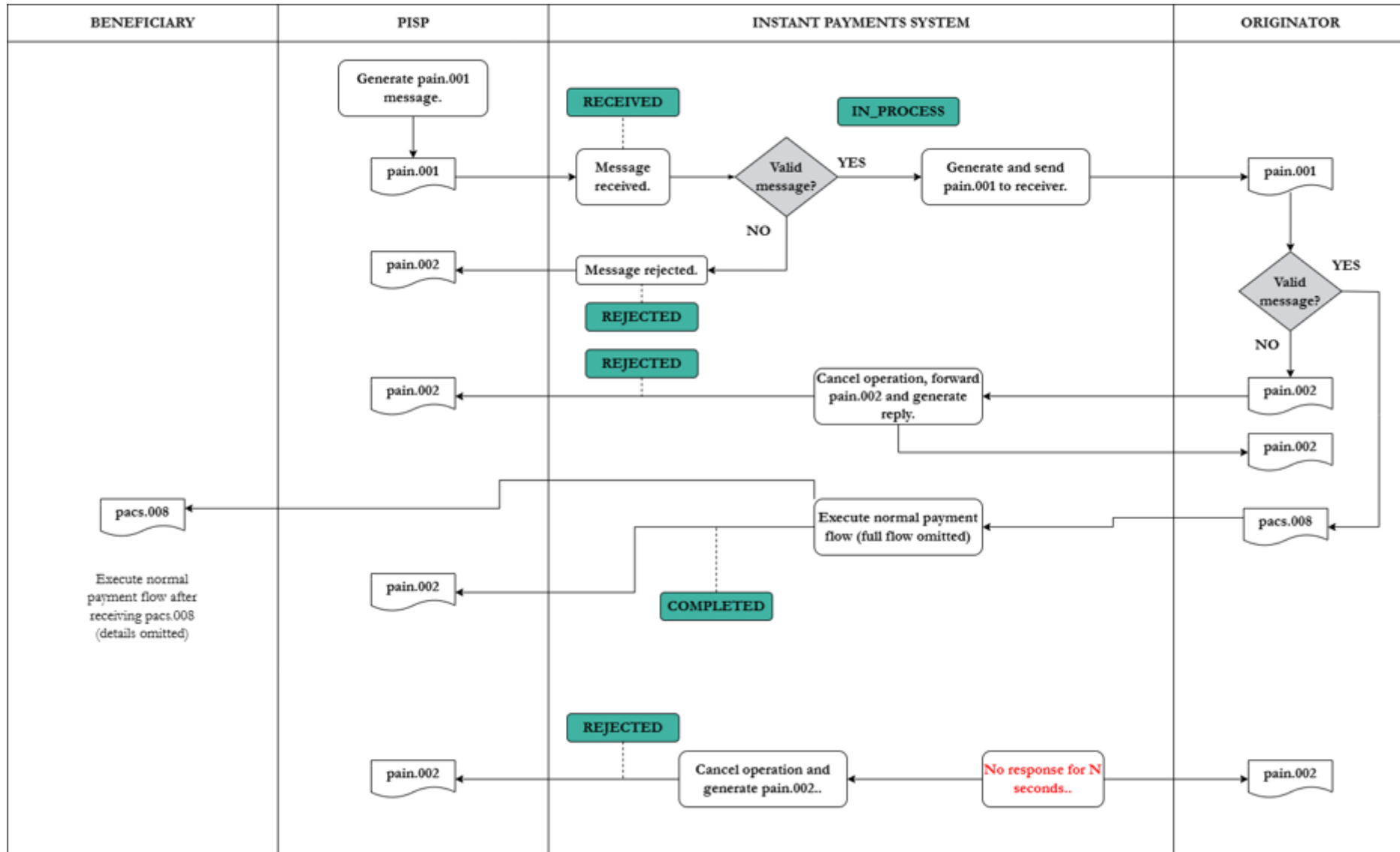


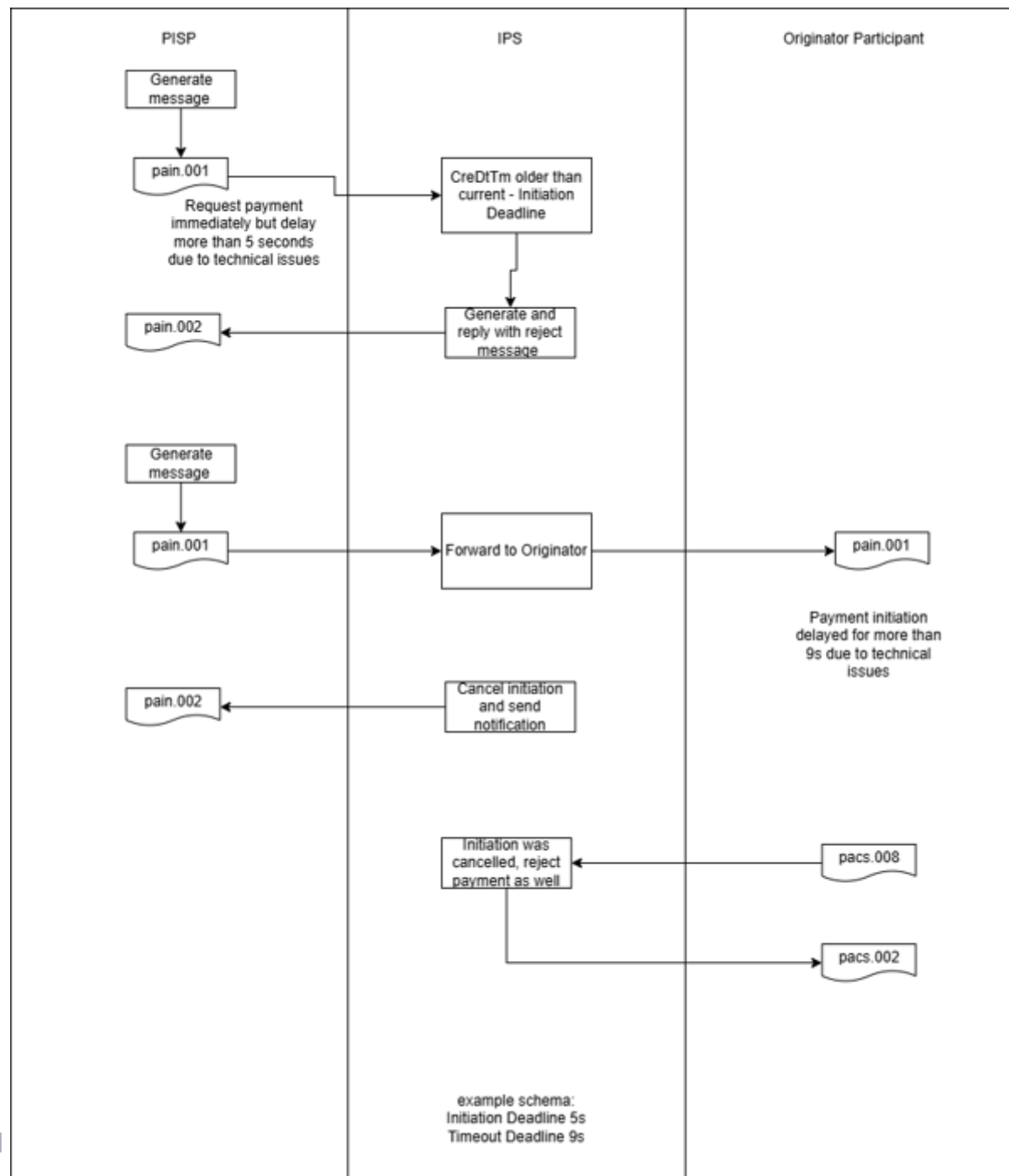


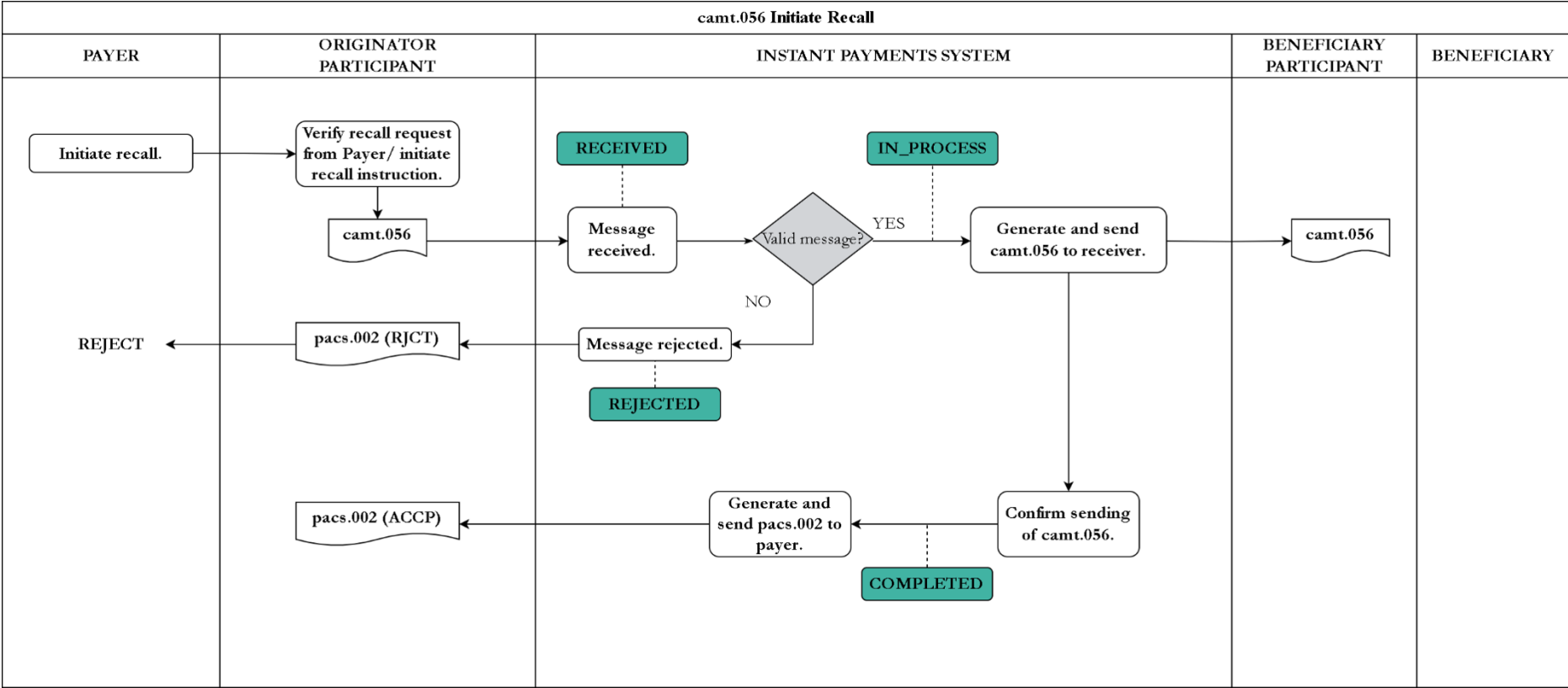


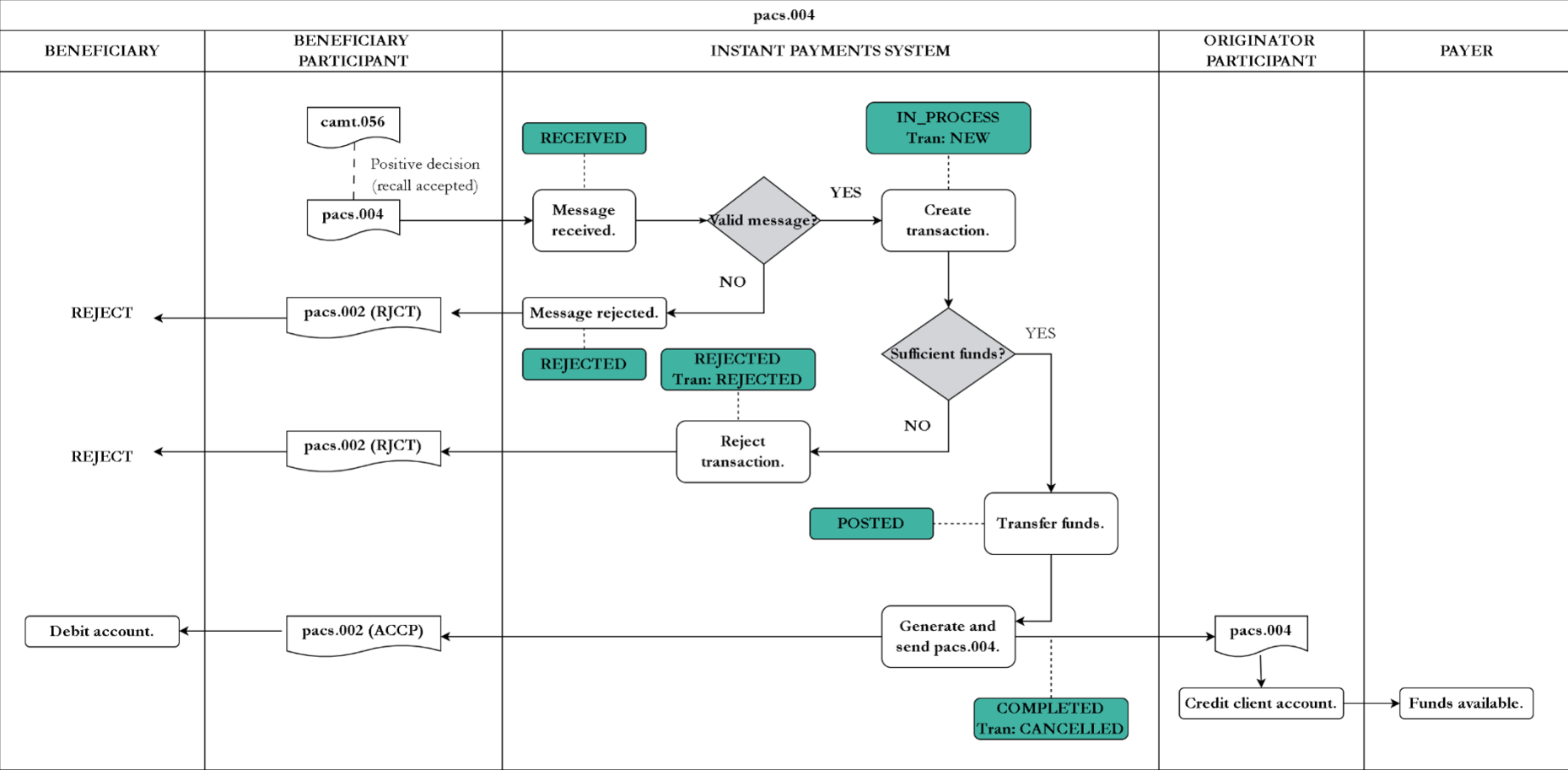


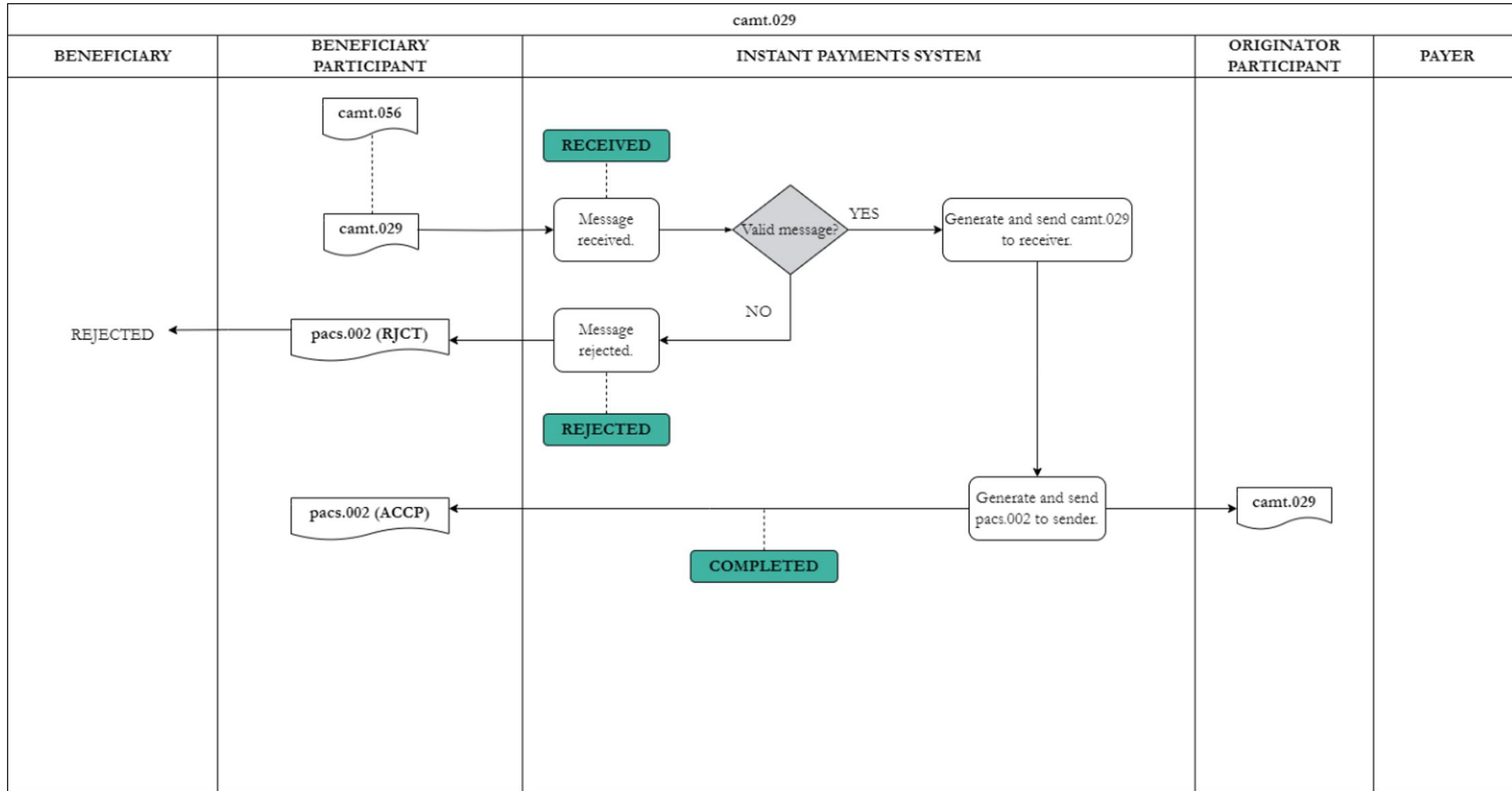


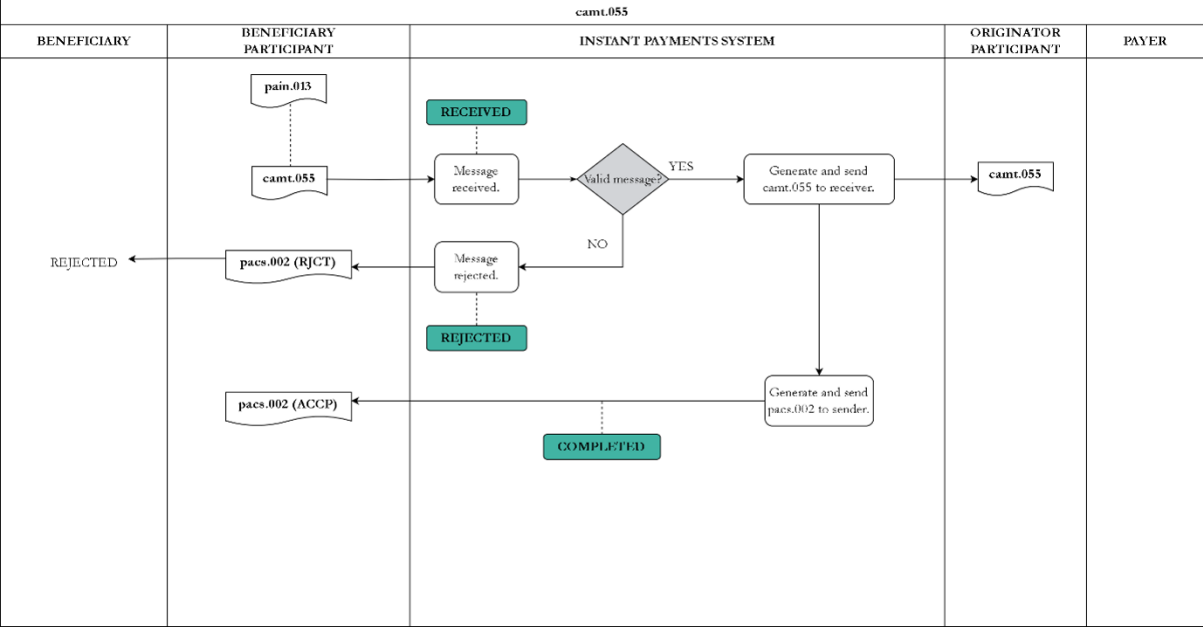


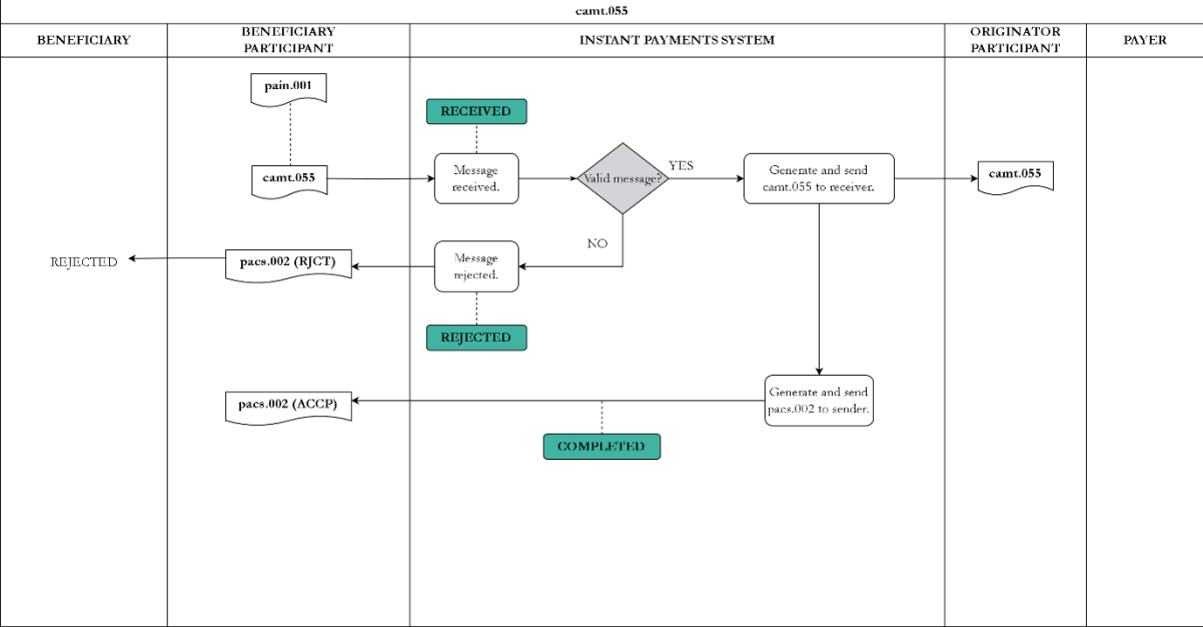


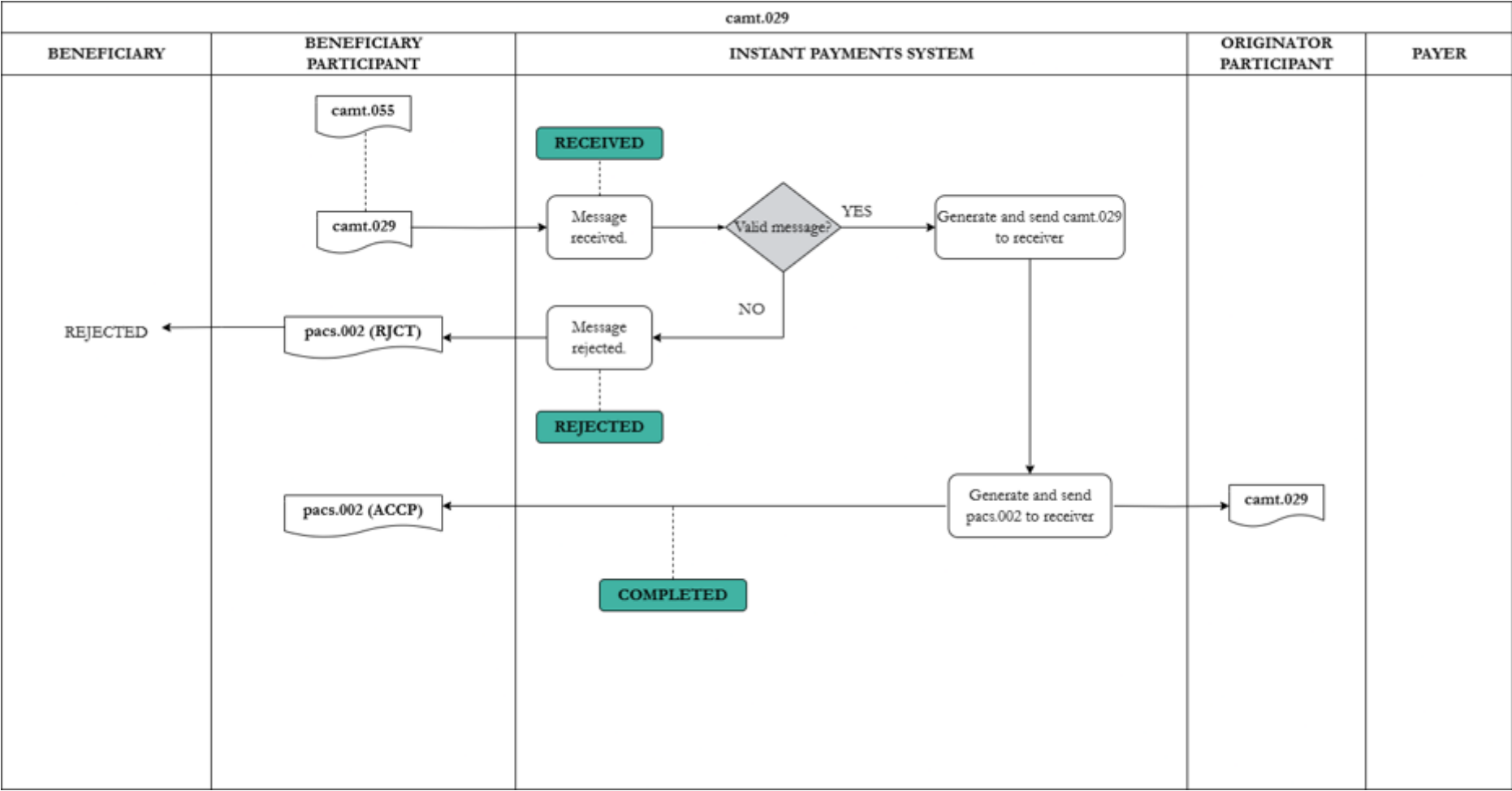


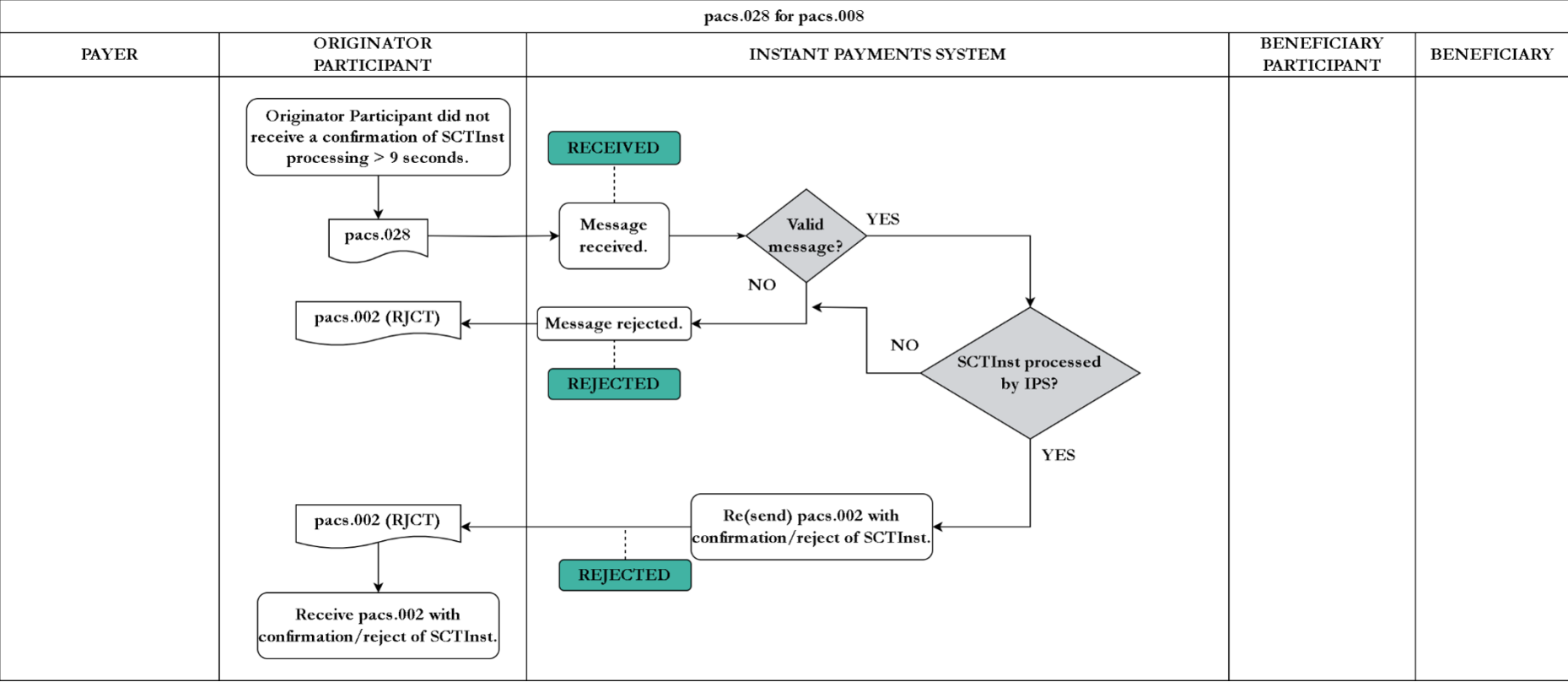


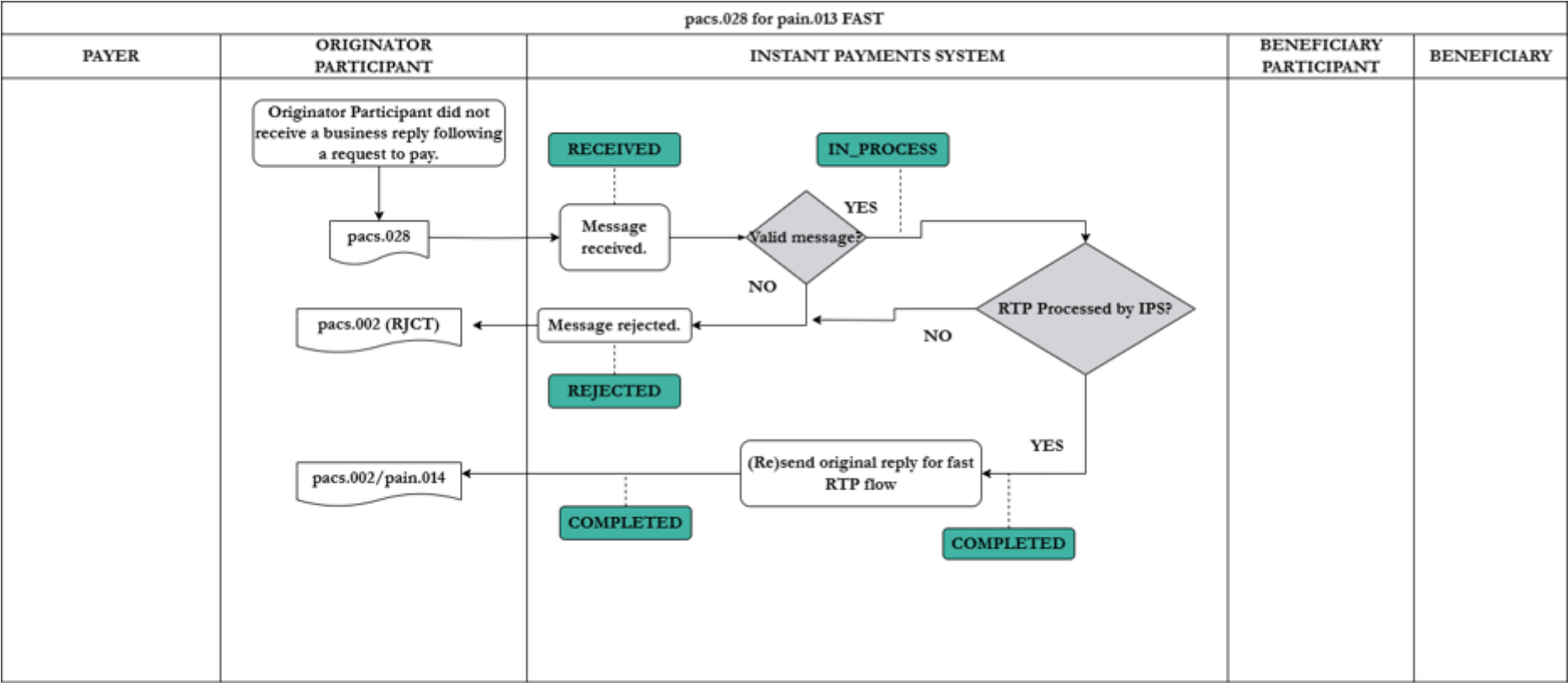


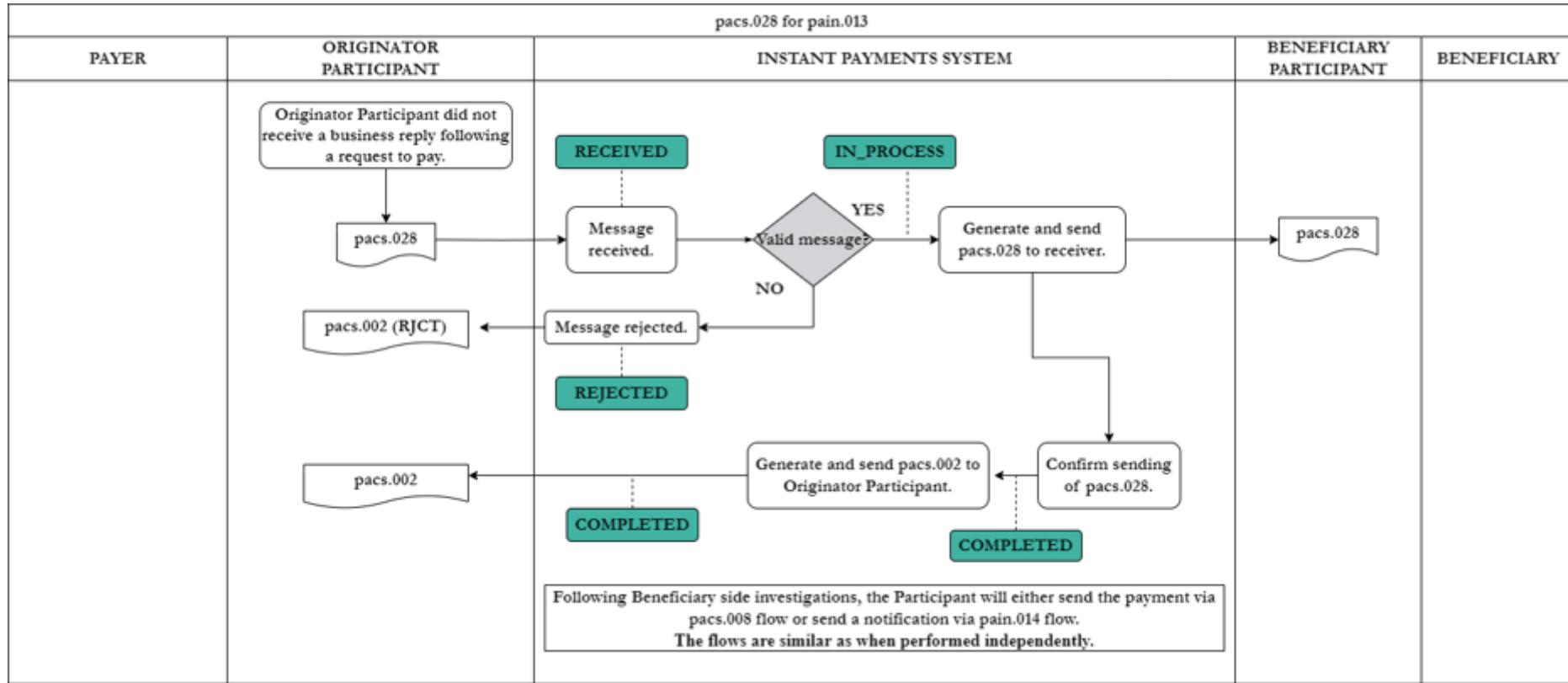


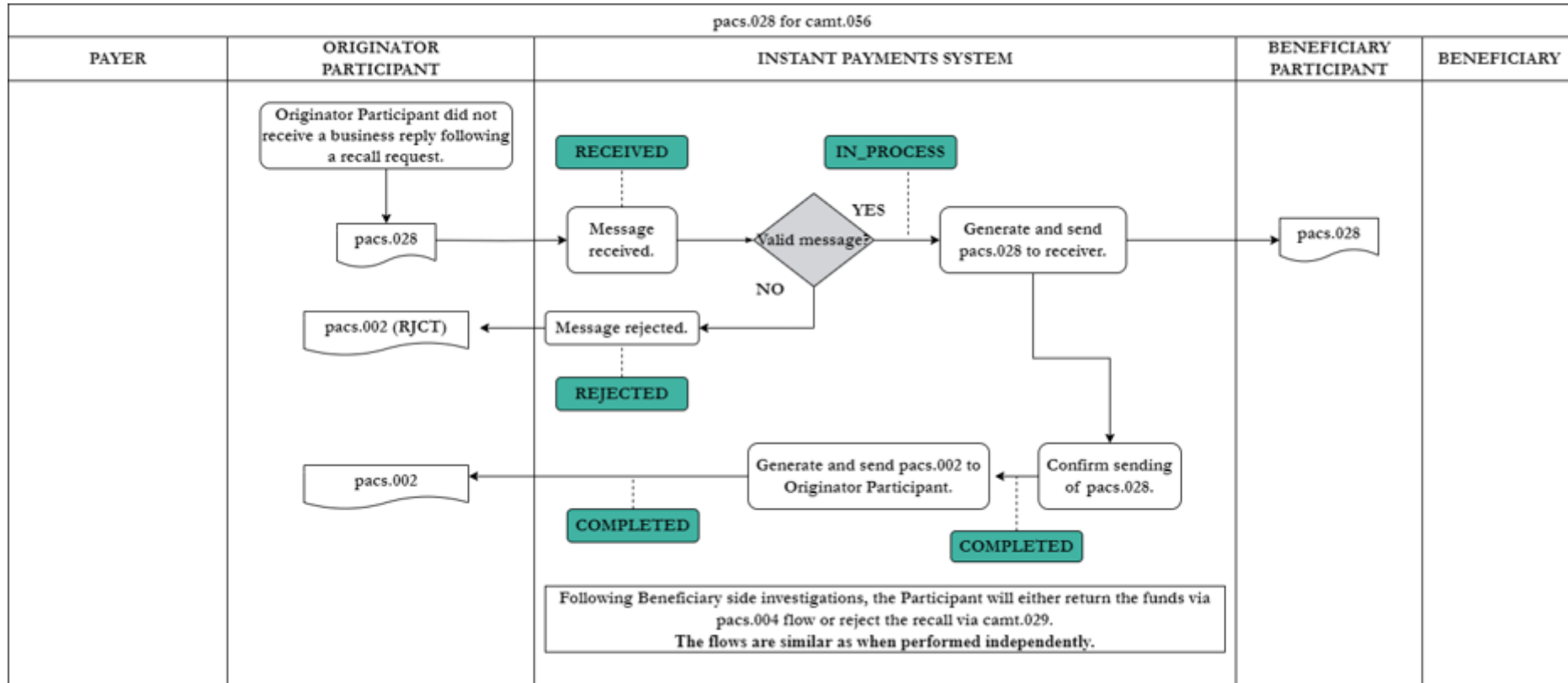












Dispute Management System

DMS

- Module for participants to help them settle disputes in IPS / RTGS / Other systems
- Designed for the participants, but Operator has full access to monitor the disputes
- Access: fully integrated in IPS MMC

DMS

■ Escalation Matrix

- » schedule containing priorities and levels - the time in which users from that specific profiles can act on the dispute
- » Levels – configurable by participants
- » Priority – fixed value, system parameter

DMS

- Disputed transactions
 - » External transactions – must be registered manually
 - » IPS transactions – validated by the system

DMS

- Creating disputes
 - » Mandatory information: receiver, data about the transaction
- Proposals
 - » Sender and receiver must agree on a proposal to solve the dispute
- Claims
 - » Generated automatically after an accepted proposal

DMS

- Correspondence
 - » Communication between sender and receiver of dispute
- Memos
 - » Communication between users belonging to same participant
- Documents
 - » Additional documents can be uploaded on the dispute

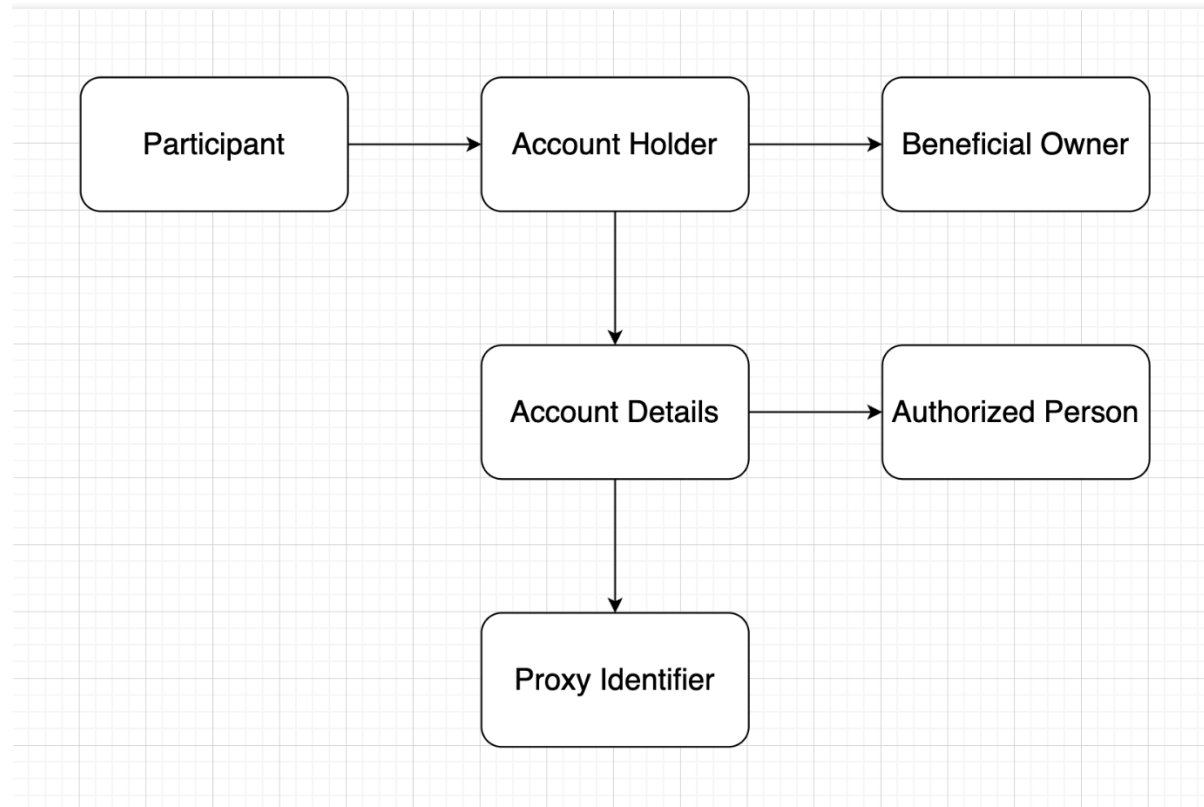
DMS

- Monitoring
 - » Dedicated dashboard for dispute summary
- Reporting
 - » Dedicated report for dispute states
- Dispute API
 - » All dispute flows are also available via API

Proxy Flows

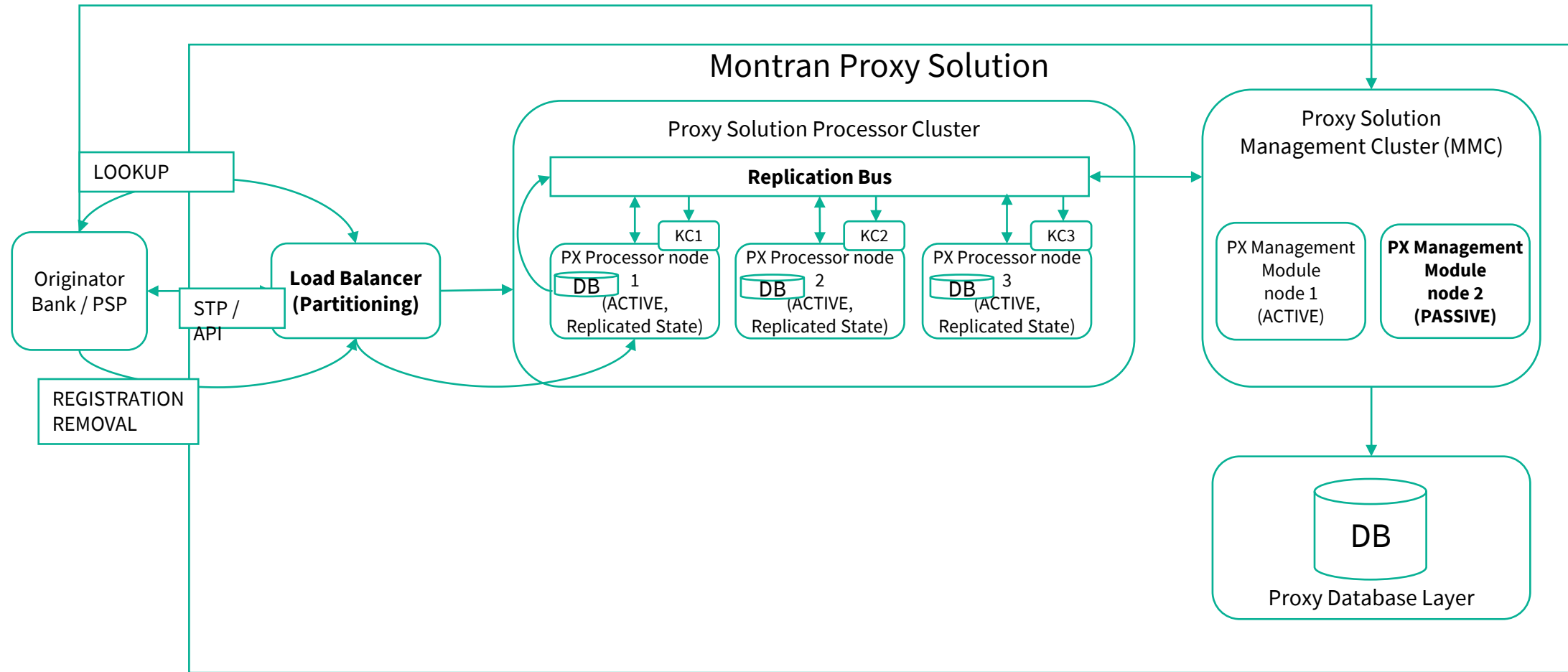
Proxy Data Types

- The Proxy Solution will handle the following data types:
 - » Account Holder
 - » Account Details
 - » Proxy Identifier
 - » Authorized Person
 - » Beneficial Owner



MONTRAN PROXY SOLUTION

Logical Architecture



Proxy Messaging – ISO 20022

■ **acmt.022**

- » Registration Operation (single + bulk)
- » Update Operation (single + bulk)
- » Removal Operation (single + bulk)

■ **acmt.023**

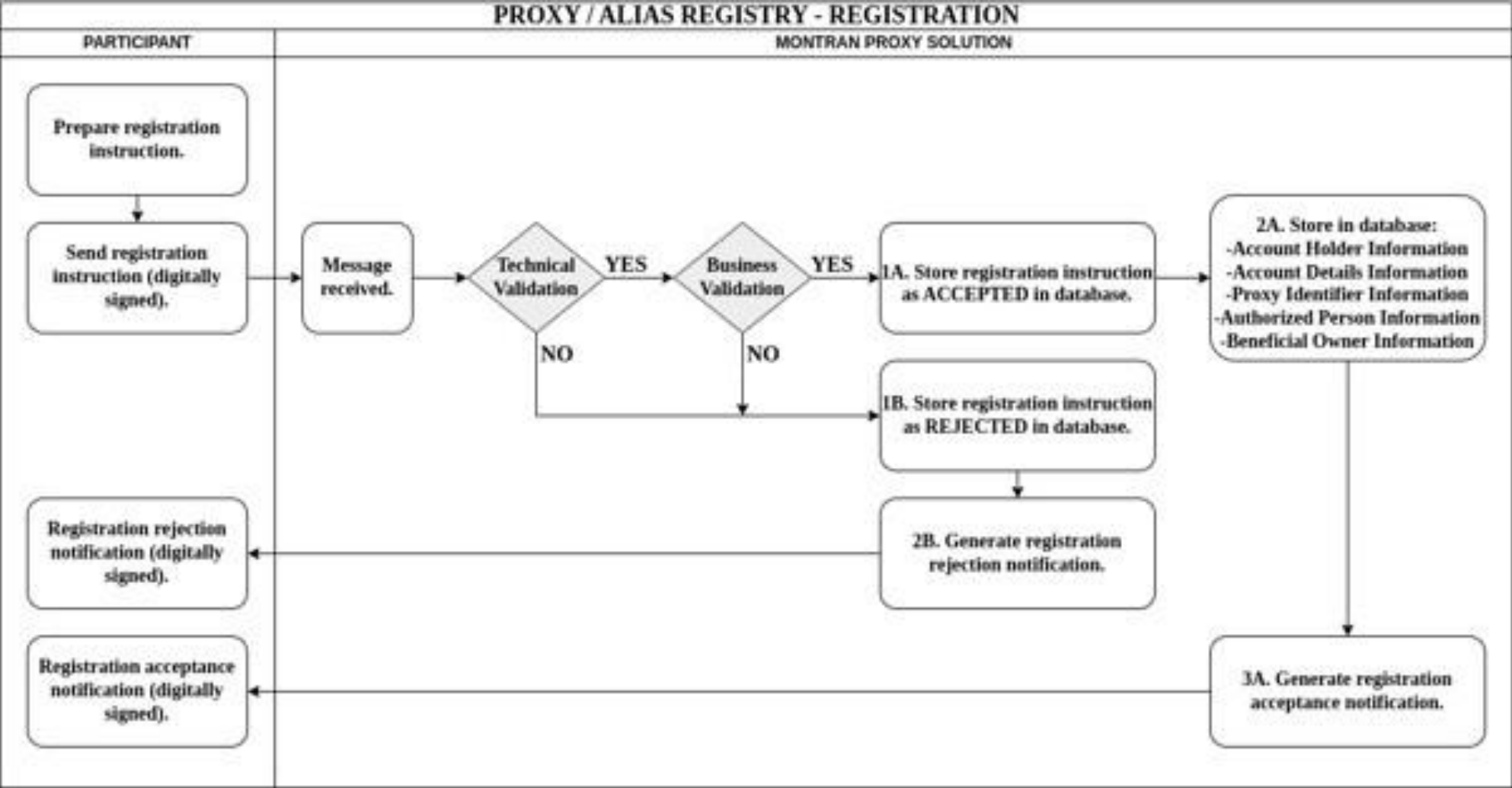
- » Proxy Lookup (single + bulk)
- » Account Inquiry (single + bulk)
- » Reachability Check (single + bulk)
- » Account Possession Check (single + bulk)
- » Verification of Payee (single + bulk)

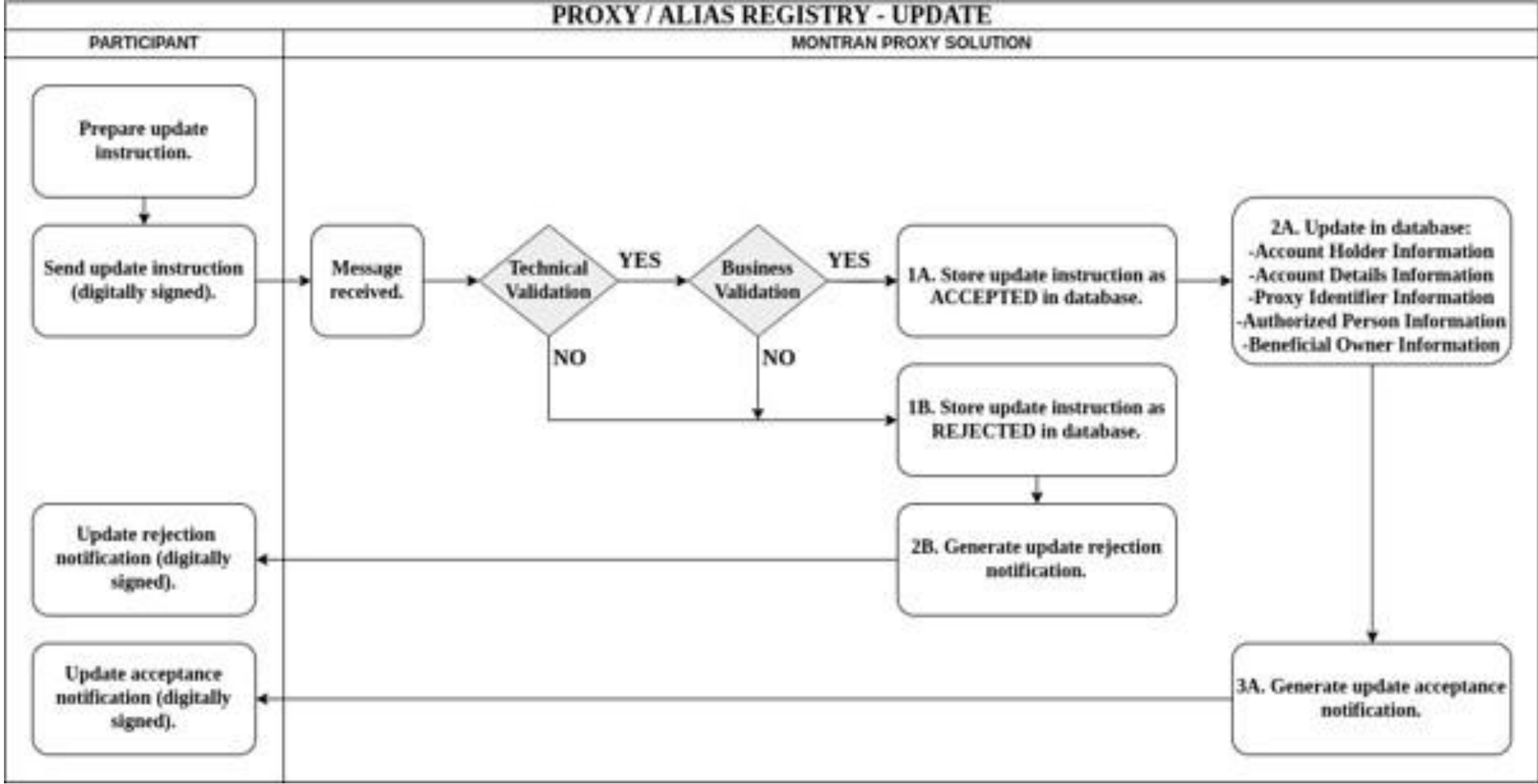
■ **Acmt.024**

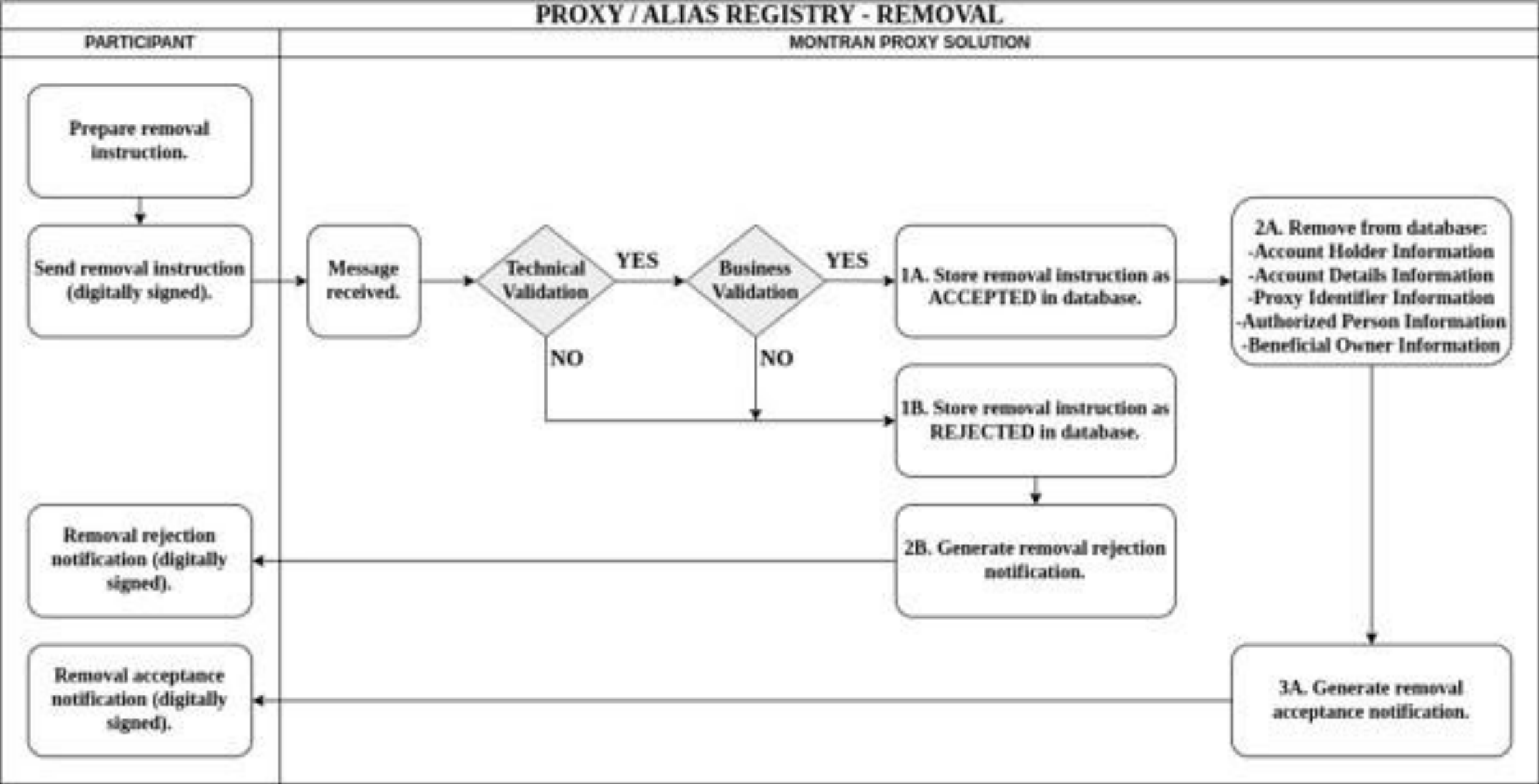
- » Response to Proxy Lookup Operation
- » Response to Account Holder Inquiry Operation
- » Response to Reachability Check Operation
- » Response to Account Possession Check Operation
- » Response to Verification of Payee Operation

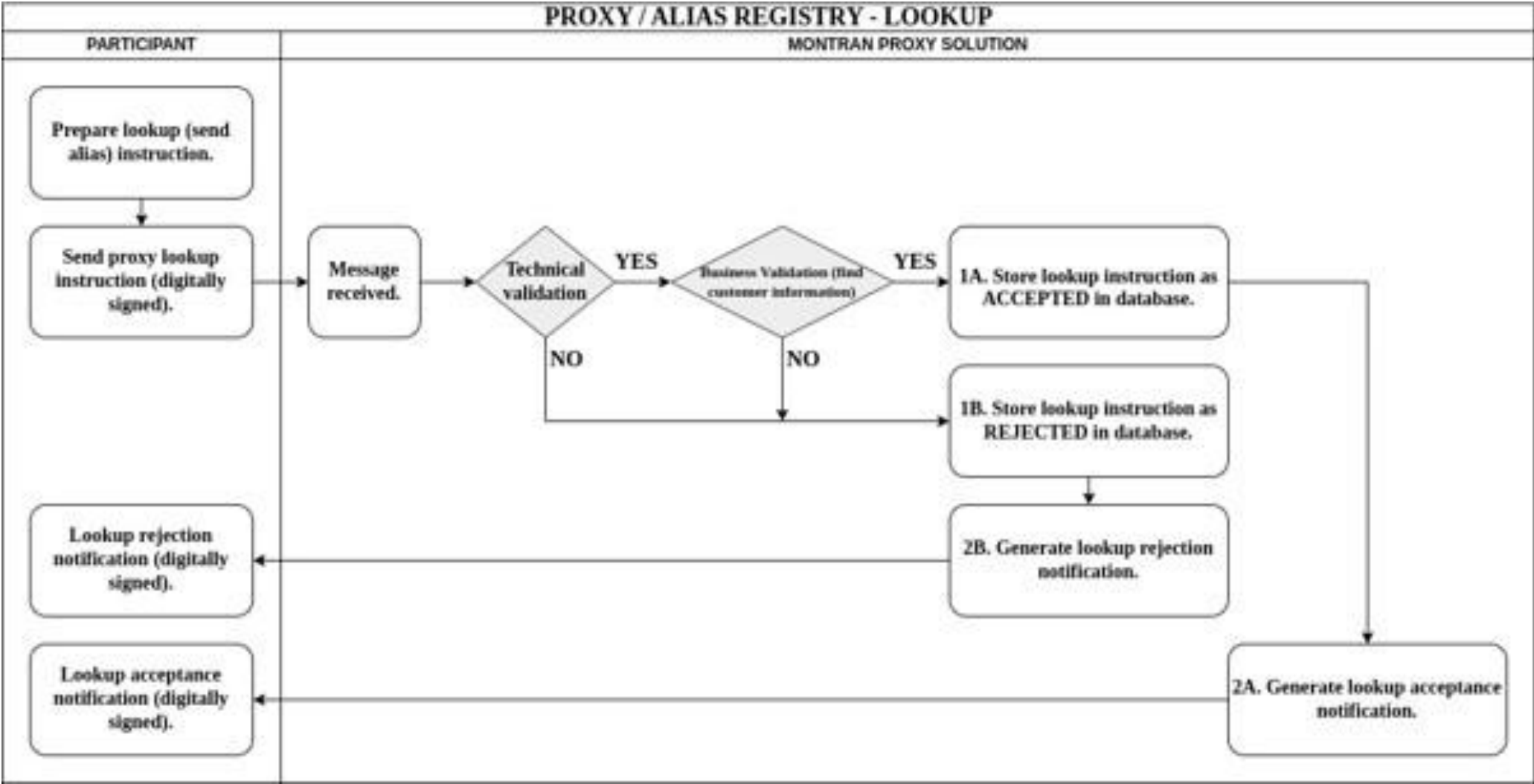
■ **Pacs.002**

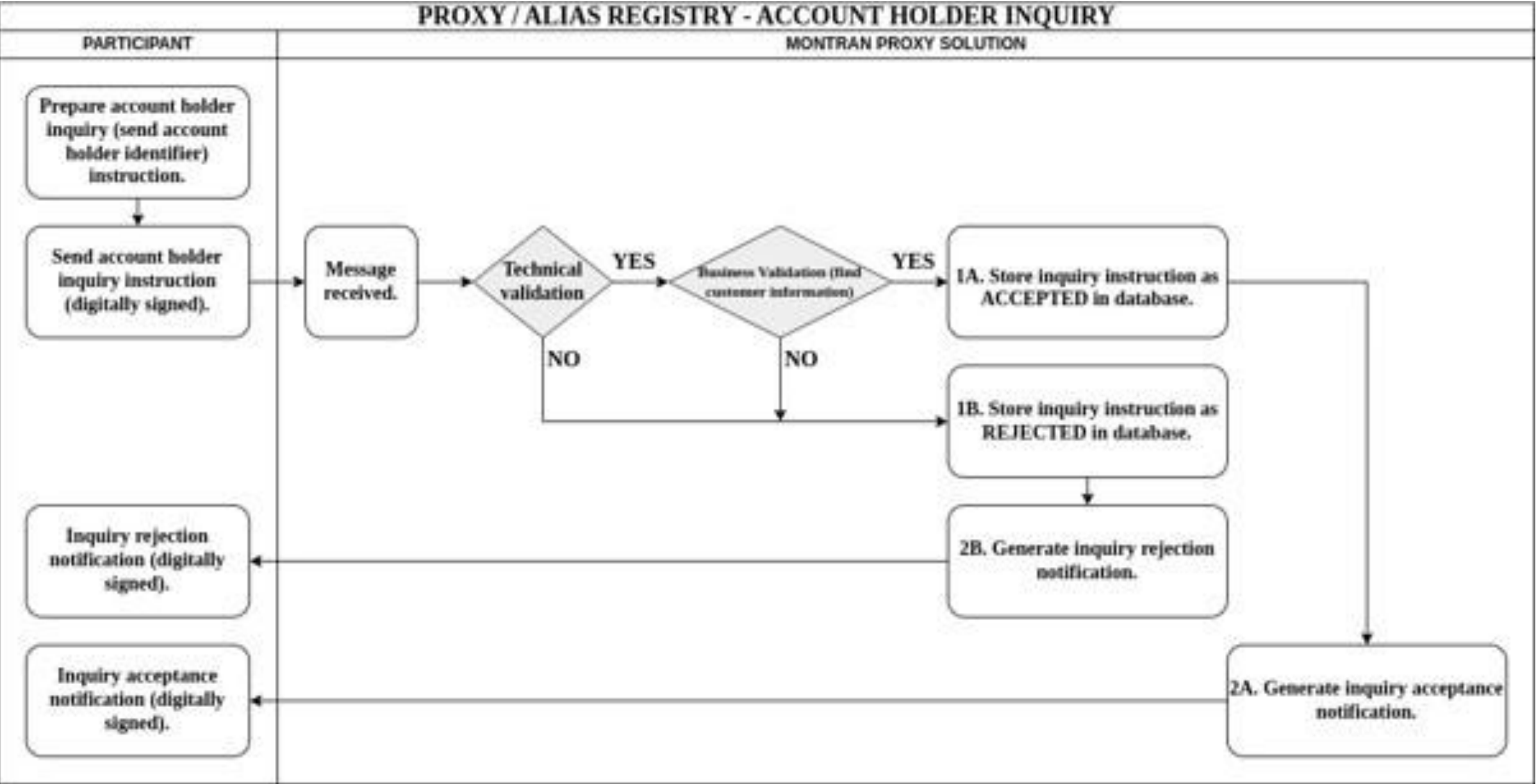
- » Response to Registration Operation
- » Response to Update Operation
- » Response to Removal Operation
- » Can also be sent as response to query operations to indicate a problem with the message (e.g., invalid xml)

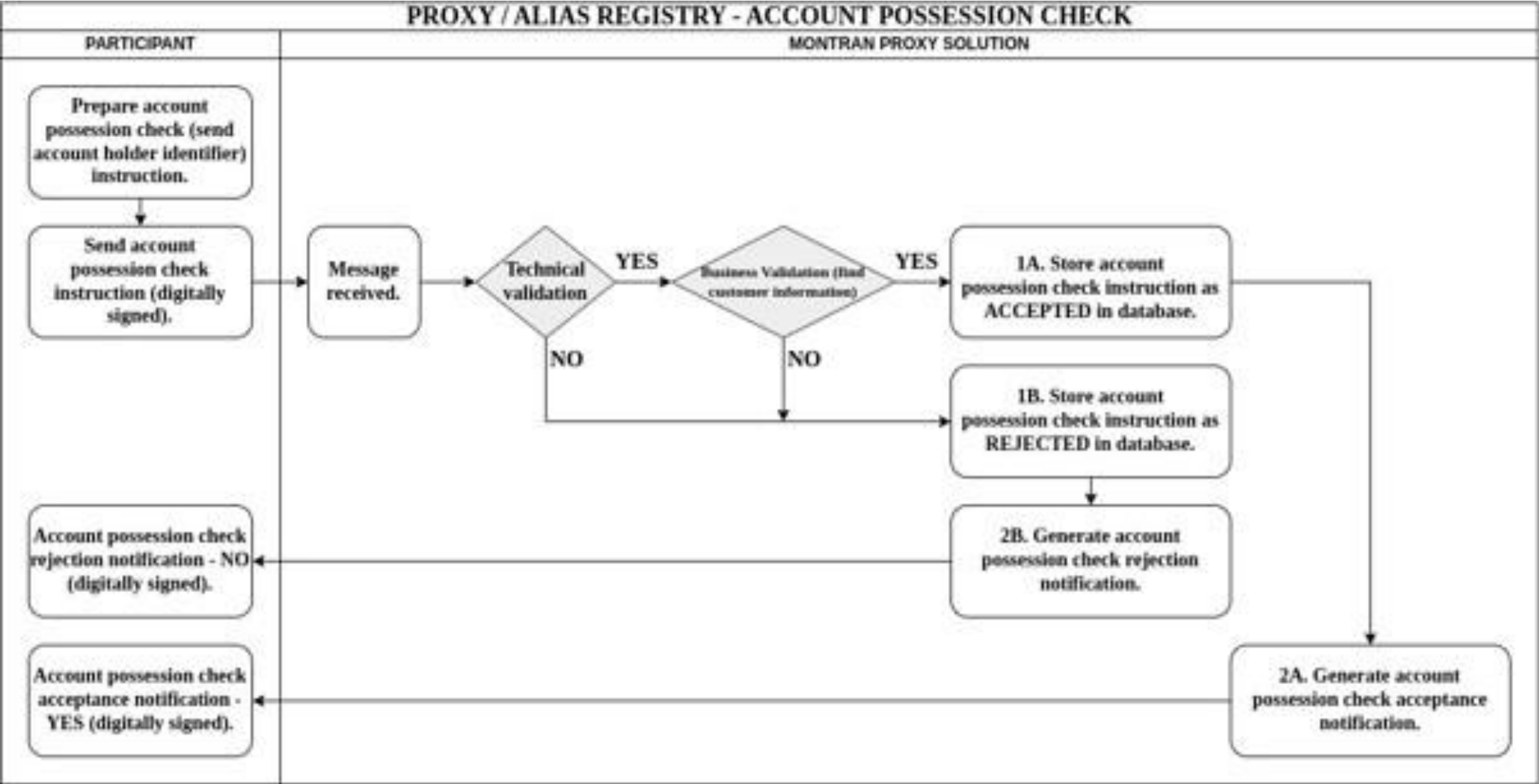


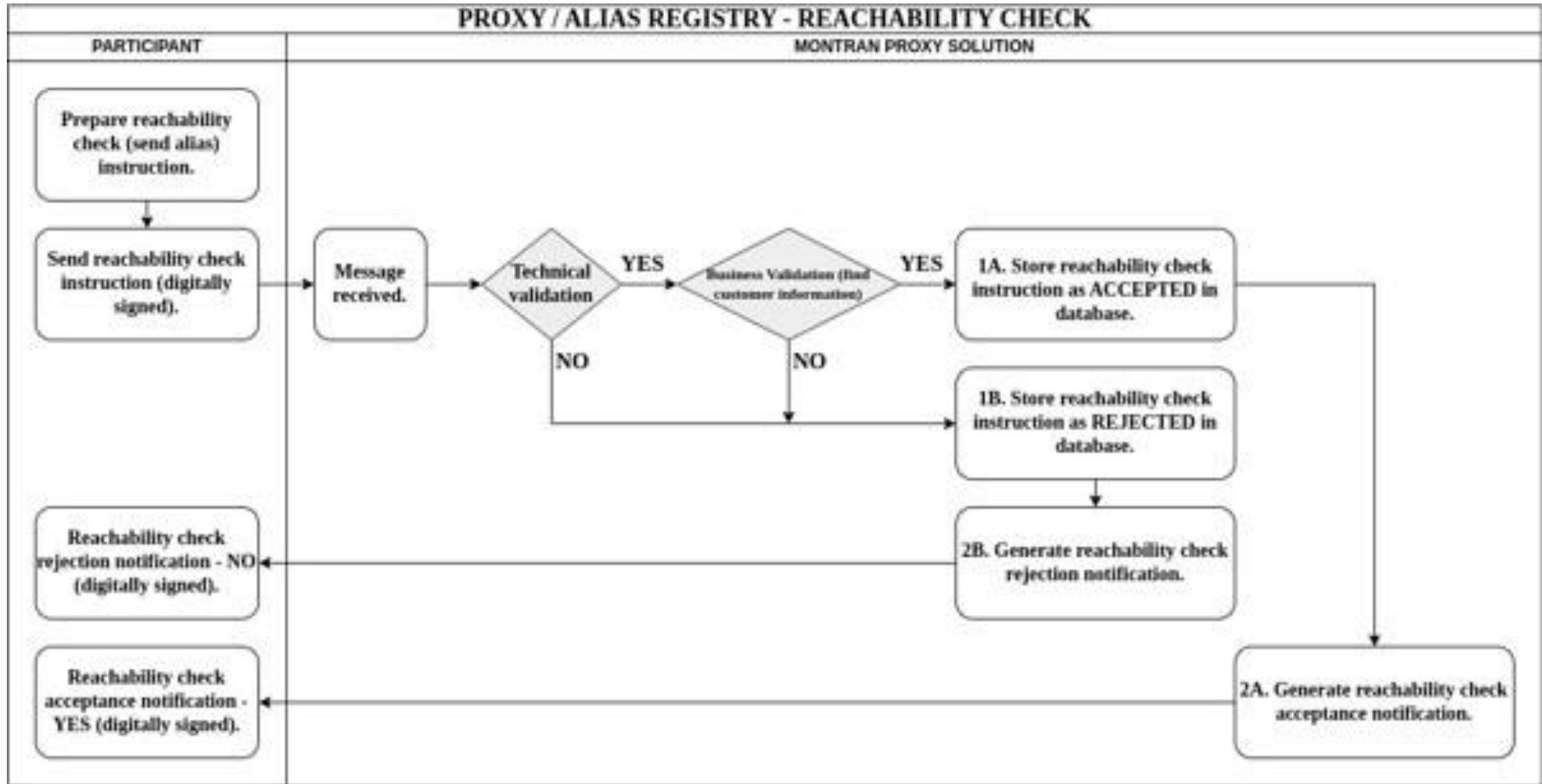


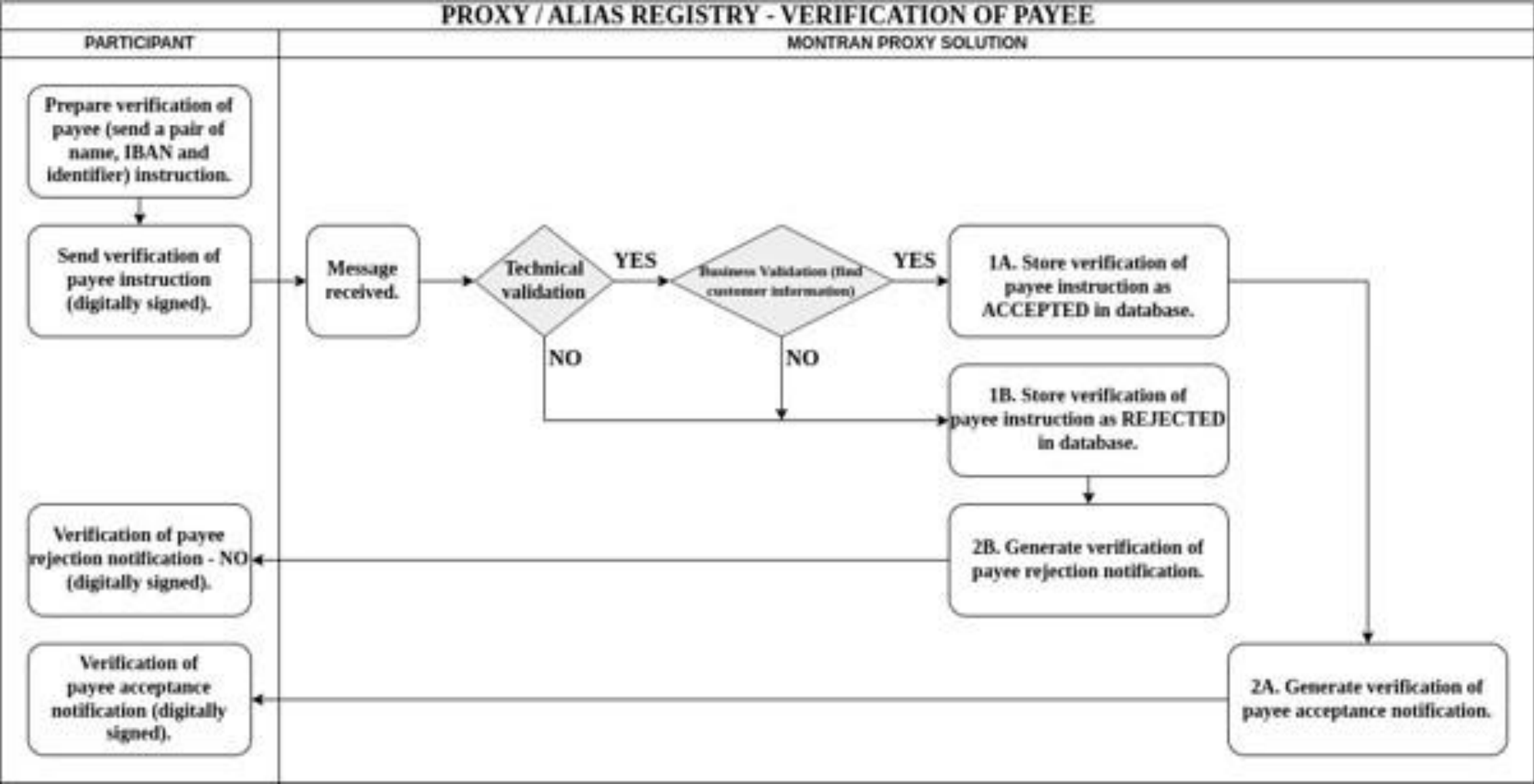












Investigation

Routing and Monitoring

- Messages: Routing → Messages → List
- Transactions: Clearing → Transactions → List
- Real-time monitoring screens: Monitoring menu
 - » Events scheduler
 - » Node status
 - » Account positions
 - » Participant status
 - » Participant message status
- Alerts: Maintenance → Alert occurrence
- Disputes Monitoring: Dispute Management → Dispute Monitoring → List

Billing

- Participant users have only list rights for the Billing menu.
- Operations available: List.
 - » Navigate to: Billing → Rate → List.
- Three main default rate categories
 - » Activity Rates
 - » Participation Rates
 - » Ad-hoc Rates.
- Additional rate types can be applied to an activity rate
 - » Volume
 - » Incentive
 - » Surcharge
- Interchange fees

Reporting

■ Generation:

- » Automatically after daily roll-over
- » Generated as set in the Timetable – Report Generation event
- » Intraday – on request

■ Reports:

- » Users Operations Journal Report
- » Current Balance Report
- » Liquidity Adjustment Report
- » Dispute Report
- » Daily Bill

- » Daily Bill Interbank Report
- » Billing Invoice
- » Billing Invoice Interbank Report
- » Reconciliation Report
- » Request to pay Report

Continue the Dialogue with Us!

PLEASE REACH OUT WITH ANY QUESTIONS